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IMPLEMENTATION COMPLETION AND RESULTS REPORT

IDA-52800 and TF-14713

ON AN

INTERNATIONAL DEVELOPMENT ASSOCIATION CREDIT

IN THE AMOUNT OF SDR 248.6 MILLION

(US\$375 MILLION EQUIVALENT)

AND

GRANT

IN THE AMOUNT OF US\$25 MILLION

FROM THE PILOT PROGRAM FOR CLIMATE RESILIENCE (PPCR)

TO THE

PEOPLE'S REPUBLIC OF BANGLADESH

FOR THE

COASTAL EMBANKMENT IMPROVEMENT PROJECT - PHASE I (CEIP-I)

June 28, 2024

CURRENCY EQUIVALENTS

(Exchange Rate Effective December 31, 2023)

Currency Unit = BDT

BDT 109.75 = US\$1

US\$ 1.34 = SDR 1

FISCAL YEAR

July 1 - June 30

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ABBREVIATIONS AND ACRONYMS

BCR	Benefit-Cost Ratio
BP	Bank Procedure
BWDB	Bangladesh Water Development Board
CEIP	Coastal Embankment Improvement Project
CEIP-I	Coastal Embankment Improvement Project - Phase I
CERC	Contingent Emergency Response Component
DDCS&PMS	Detailed Design Construction Supervision and Project Management Support
DPP	Development Project Proforma
EA	Environmental Assessment
EIA	Environmental Impact Assessment
EIRR	Economic Internal Rate of Return
EMP	Environmental Management Plan
FAPAD	Foreign Aided Project Directorate
FM	Financial Management
GoB	Government of Bangladesh
GRC	Grievance Redress Committee
GRM	Grievance Redress Mechanism
IRI	Intermediate Results Indicator
LAP	Land Acquisition Plan
M&E	Monitoring & Evaluation
MTR	Mid-Term Review
NPV	Net Present Value
O&M	Operation and Maintenance
OP	Operational Policy
PAD	Project Appraisal Document
PDO	Project Development Objectives
PMU	Project Management Unit
PPCR	Pilot Program for Climate Resilience
RAP	Resettlement Action Plan
SLR	Sea Level Rise
SMRPF	Social Management and Resettlement Policy Framework
VSL	Value of a Statistical Life
WMA	Water Management Association
WMG	Water Management Group
WMO	Water Management Organization

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**DATA SHEET****BASIC INFORMATION****Product Information**

Project ID	Project Name
P128276	Coastal Embankment Improvement Project - Phase I (CEIP-I)
Country	Financing Instrument
Bangladesh	Investment Project Financing
Original EA Category	Revised EA Category
Full Assessment (A)	Full Assessment (A)

Organizations

Borrower	Implementing Agency
People's Republic of Bangladesh	Bangladesh Water Development Board

Project Development Objective (PDO)

Original PDO

The project development objectives are to (a) increase the area protected in selected polders from tidal flooding and frequent storm surges, which are expected to worsen due to climate change; (b) improve agricultural production by reducing saline water intrusion in selected polders; and (c) improve the Government of Bangladesh's capacity to respond promptly and effectively to an eligible crisis or emergency.



FINANCING

	Original Amount (US\$)	Revised Amount (US\$)	Actual Disbursed (US\$)
World Bank Financing			
IDA-52800	375,000,000	350,740,001	311,840,007
TF-14713	25,000,000	24,999,892	24,999,892
Total	400,000,000	375,739,893	336,839,899
Non-World Bank Financing			
Borrower/Recipient	0	0	0
Total	0	0	0
Total Project Cost	400,000,000	375,739,893	336,839,899

KEY DATES

Approval	Effectiveness	MTR Review	Original Closing	Actual Closing
26-Jun-2013	24-Nov-2013	13-Jan-2017	31-Dec-2020	31-Dec-2023

RESTRUCTURING AND/OR ADDITIONAL FINANCING

Date(s)	Amount Disbursed (US\$M)	Key Revisions
12-Jan-2020	203.77	Change in Results Framework Change in Loan Closing Date(s) Change in Implementation Schedule
06-Mar-2022	266.82	Change in Loan Closing Date(s)
12-Dec-2023	336.84	Change in Components and Cost Cancellation of Financing Reallocation between Disbursement Categories

KEY RATINGS

Outcome	Bank Performance	M&E Quality
Moderately Satisfactory	Moderately Satisfactory	Substantial

**RATINGS OF PROJECT PERFORMANCE IN ISRs**

No.	Date ISR Archived	DO Rating	IP Rating	Actual Disbursements (US\$M)
01	29-Oct-2013	Satisfactory	Satisfactory	0
02	25-Jun-2014	Satisfactory	Moderately Satisfactory	4.63
03	22-Dec-2014	Satisfactory	Satisfactory	4.63
04	15-Jun-2015	Satisfactory	Satisfactory	4.63
05	24-Dec-2015	Satisfactory	Satisfactory	26.20
06	13-Jul-2016	Satisfactory	Moderately Satisfactory	26.20
07	06-Mar-2017	Satisfactory	Moderately Satisfactory	41.50
08	30-Oct-2017	Satisfactory	Moderately Unsatisfactory	72.97
09	09-May-2018	Satisfactory	Moderately Unsatisfactory	72.97
10	13-Nov-2018	Satisfactory	Moderately Satisfactory	141.43
11	01-May-2019	Satisfactory	Moderately Satisfactory	161.21
12	11-Nov-2019	Satisfactory	Moderately Satisfactory	191.11
13	16-Jun-2020	Satisfactory	Moderately Satisfactory	235.81
14	07-Oct-2020	Satisfactory	Moderately Satisfactory	235.81
15	10-Apr-2021	Satisfactory	Moderately Satisfactory	245.84
16	07-Oct-2021	Satisfactory	Moderately Satisfactory	261.83
17	06-Apr-2022	Satisfactory	Satisfactory	274.82
18	08-Oct-2022	Satisfactory	Satisfactory	280.48
19	26-Apr-2023	Satisfactory	Moderately Satisfactory	297.95
20	18-Dec-2023	Satisfactory	Moderately Satisfactory	336.84

**SECTORS AND THEMES****Sectors**

Major Sector/Sector	(%)
Agriculture, Fishing and Forestry	1
Forestry	1
Public Administration	12
Other Public Administration	12
Social Protection	13
Social Protection	13
Transportation	12
Other Transportation	12
Water, Sanitation and Waste Management	62
Other Water Supply, Sanitation and Waste Management	62

Themes

Major Theme/ Theme (Level 2)/ Theme (Level 3)	(%)
Finance	11
Finance for Development	11
Disaster Risk Finance	11



Social Development and Protection	9
Social Inclusion	8
Other Excluded Groups	7
Participation and Civic Engagement	1
Fragility, Conflict and Violence	1
Forced Displacement	1
Urban and Rural Development	33
Disaster Risk Management	33
Disaster Response and Recovery	11
Disaster Risk Reduction	11
Disaster Preparedness	11
Environment and Natural Resource Management	118
Climate change	71
Adaptation	71
Environmental policies and institutions	2
Water Resource Management	45
Water Institutions, Policies and Reform	45

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I. PROJECT CONTEXT AND DEVELOPMENT OBJECTIVES

A. CONTEXT AT APPRAISAL

Context

1. **At appraisal, Bangladesh's vulnerability to the effects of climate change threatened its potential for future growth, and this continues today.** The country's low-lying topography, prone to floods, cyclonic storms, and erosion, posed significant development challenges, particularly in its coastal zones. Hosting nearly 28 percent of the country's population, the coastal zones of Bangladesh were characterized by rapid population growth, poverty, and low elevation, making them susceptible to multiple threats. It was estimated that the coastal population would grow to 61 million by 2050, further escalating the population density in these low-lying regions. Food self-sufficiency was a major development goal, given that a significant portion of the country's poor population relied heavily on agriculture for income and spent a large part of their earnings on food. Consequently, increased vulnerability in coastal farming had negative impacts on the welfare of farmers and poorer consumers across the country.

2. **The Government of Bangladesh (GoB) made significant investments and policy changes to protect its coastal population from natural hazards.** It invested approximately US\$10 billion in structural and non-structural disaster mitigation and preparedness systems, including embankments, polders, cyclone shelters, and early warning systems, resulting in the establishment of over 2,130 cyclone shelters, 139 polders, and 2,900 water control structures at the time of appraisal. The GoB also created a supportive policy environment for mainstreaming coastal development, water resource management, disaster risk management, and climate change adaptation agendas in its major development strategies.

3. **Despite the enhanced resilience provided by these policies and interventions, the coastal population's vulnerability was on the rise because of climate change.** Climate variability exacerbated the inherent risks of cyclones, storm surges, riverbank erosion, sea level rise (SLR), saline intrusion, and coastal erosion. A study found that at the time of appraisal, a 10-year return period cyclone scenario would affect 8 million people with inundation depths of over 3 meters; by 2050, the number affected would rise to 22.5 million people as a result of population growth and climate change. It identified vulnerable polders and emphasized the cost effectiveness of investing in adaptation measures.¹ The existing coastal embankments, which provided protection against these hazards, required retrofitting and upgrading to maintain their effectiveness. The integrity of the embankments was further threatened by the siltation of peripheral rivers, poor maintenance, and inadequate management. The effectiveness of embankments in reducing damages and saving lives during Cyclone Sidr prompted extensive consultations between government agencies, development partners, and the World Bank to prepare a long-term disaster risk reduction program, which included the construction of multipurpose cyclone shelters and the rehabilitation and upgrading of the embankment system as key climate adaptation investments.

4. **The Bangladesh Water Development Board (BWDB), with its extensive experience in managing coastal embankment systems and implementing various projects, played a vital role in the country's water resource management.** Operational since 1959, it executed flood, drainage, and irrigation plans; collected and disseminated hydrologic and hydraulic data; and managed the Flood Forecasting and Warning Center. It had successfully implemented the IDA-financed Coastal Embankment Rehabilitation Project and at the time of appraisal, was rehabilitating

¹ Adams, Norma, Susmita Dasgupta, and Maria Sarraf. 2011. *The Cost of Adapting to Extreme Weather Events in a Changing Climate (English)*. Bangladesh Development Series, Paper No. 28. Washington, DC: World Bank Group.



embankments destroyed by Cyclone Sidr under the Emergency 2007 Cyclone Recovery and Restoration Project. However, the BWDB faced governance risks, particularly with respect to contract management and procurement, since it did not have adequate procurement capacity to manage high-value contracts. It did have relatively good financial management (FM) capacity in terms of having a robust FM system with an operational computerized accounting process.

5. The Coastal Embankment Improvement Project (CEIP) - Phase I (CEIP-I) utilized advanced modeling techniques to design a robust coastal polder system for the first time in history in Bangladesh. The project used a hydrodynamic cyclone and storm surge model to simulate 19 major cyclones from 1960 to 2009, considering various factors such as tidal conditions, climate change, and SLR. The model results informed the design of embankments, slope protection, and drainage systems. A validated drainage model was used to assess the performance of the existing and future drainage networks, ensuring effective drainage of rainfall storms with a 10-year frequency. The project also accounted for land subsidence and adopted a 0.5 m SLR by 2050, based on the Intergovernmental Panel on Climate Change's 2007 *Fourth Assessment Report*² and local measurements.

6. The project supported the national development priorities of the GoB and contributed to the World Bank's Country Assistance Strategy for FY2011–14. The Sixth Five Year Plan (FY2011–15) of the GoB acknowledged the importance of supporting rural communities to strengthen their resilience and adapt to climate change, emphasizing the need to invest in flood management schemes, and repair and maintain existing coastal polders to protect and enhance agricultural productivity. The project's objectives also remained highly relevant to the development priorities of the GoB outlined in the Seventh Five Year Plan (FY2016–20). The plan's approach and goals for disaster risk management with the rehabilitation of embankments can be considerably attributed to the foundation laid by CEIP-I activities. The project aligned with the second pillar of the World Bank's Country Assistance Strategy for FY2011–14, which aimed to reduce environmental degradation and vulnerability to climate change and natural disasters by enhancing the performance of coastal embankments. In addition, the project's interventions remained highly relevant for the Country Partnership Framework (CPF) (FY2015–22), which outlined climate and environmental management as one of its three pillars.³ Furthermore, the project was aligned with the Pilot Program for Climate Resilience (PPCR) under the Climate Investment Fund by increasing the resilience of coastal infrastructure, reducing water and soil salinity, and improving agricultural production.

Bank Rationale for Engagement

7. The World Bank played (and continues to play) a critical role in strengthening disaster risk management in coastal areas of Bangladesh through various projects and initiatives. Following Cyclone Sidr in 2007, the Bank supported the GoB in the construction and rehabilitation of disaster shelters and coastal embankments through the Emergency 2007 Cyclone Recovery and Restoration Project (P111272) and the Multipurpose Disaster Shelter Project (P146464). The ongoing Resilient Infrastructure for Adaptation and Vulnerability Reduction Project (P173312) aims to reduce vulnerability to riverine and flash floods and improve disaster preparedness and response. In CEIP-I, the World Bank could build on its successes with similar projects in Bangladesh and other countries to develop scalable models of polders to address the frequent and recurrent tidal and storm surges exacerbated by climate change. The World Bank's strong engagement in disaster risk management and climate change, along with its convening abilities, positioned it well to support the GoB in this effort.

² See <https://www.ipcc.ch/assessment-report/ar4/>.

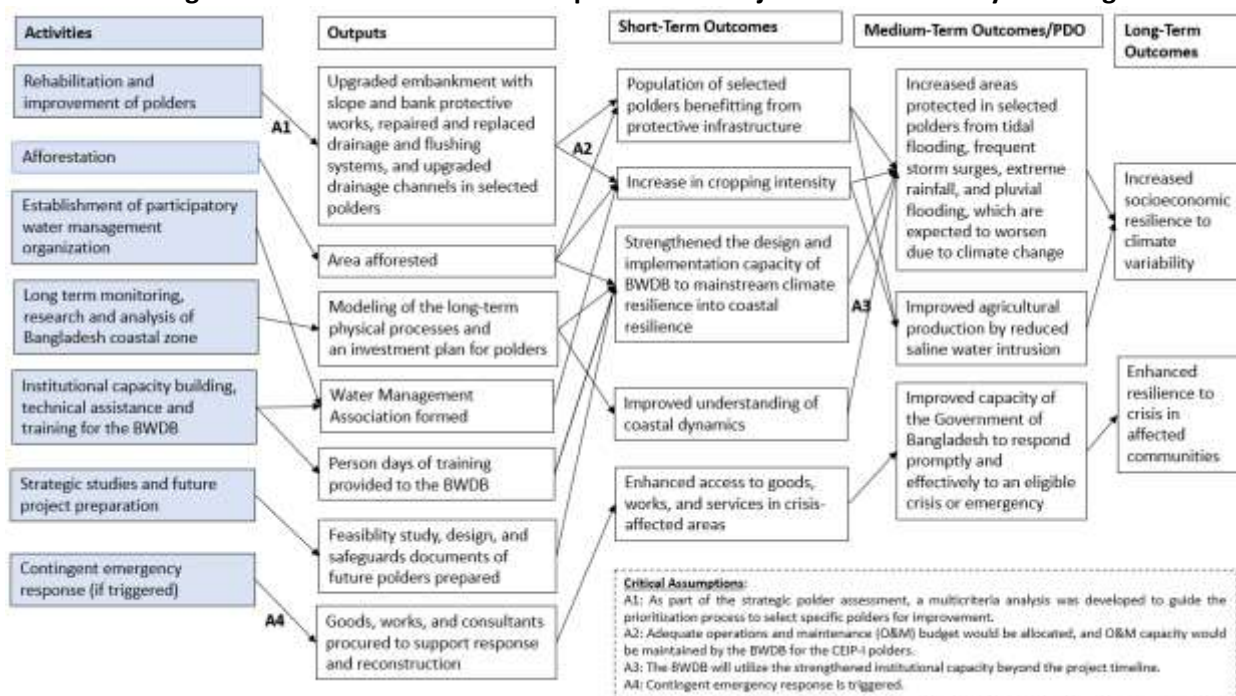
³ Under this pillar, World Bank Group activities focused on boosting Bangladesh's resilience to climate change and natural disasters; improving the management of water infrastructure; and promoting agricultural productivity with climate-smart farm practices and technology, diversification, and accelerating the move up the value chain.



Theory of Change (Results Chain)

8. The project's Theory of Change, described in Figure 1, is derived from the project context, higher-level objectives, outcomes, components, and activities described in the project appraisal document (PAD).

Figure 1: Coastal Embankment Improvement Project - Phase I Theory of Change



Project Development Objectives

9. According to the PAD and the Financing Agreement, the Project Development Objectives (PDOs) of CEIP-I were to “(a) increase the area protected in selected polders from tidal flooding and frequent storm surges, which are expected to worsen due to climate change; (b) improve agricultural production by reducing saline water intrusion in selected polders; and (c) improve the Government of Bangladesh’s capacity to respond promptly and effectively to an eligible crisis or emergency.” The targeted polders noted in the PDOs were selected through a multicriteria analysis, which relied on the following key criteria to guide the prioritization process: the physical condition of the embankment and the drainage system, economic activities in the polders (agriculture, fishery, or forestry), population and socioeconomic conditions, environmental conditions, and economic efficiency considerations (such as that the geographic proximity of polders can facilitate the efficient execution of works). CEIP-I was implemented by the BWDB with an IDA credit of US\$375 million and PPCR grant financing of US\$25 million.

Key Expected Outcomes and Outcome Indicators

10. The key expected outcomes and PDO-level indicators used to measure them are presented in Table 1.



Table 1: Project Outcomes and PDO-level Indicators

Project Outcome	PDO-level Indicator
Outcome 1: Increased area protected in selected polders from tidal flooding and frequent storm surges, which are expected to worsen due to climate change	- Gross area protected (1000 x ha) - Direct project beneficiaries from increased resilience to climate change (1000 x person), of which female (percentage)
Outcome 2: Improved agricultural production by reducing saline water intrusion in selected polders	Increase cropping intensity (percentage)
Outcome 3: Improved capacity of the Government of Bangladesh to respond promptly and effectively to an eligible crisis or emergency	Contingent emergency appropriation

Source: The World Bank. CEIP-I PAD.

Components

11. Component A – Rehabilitation and Improvement of Polders (original cost: US\$291.00 million; actual cost: US\$215.56 million)

a. **Subcomponent A1: Rehabilitation and Improvement of Polders.** The objective of this component was to increase community resilience to tidal flooding and storm surges by rehabilitating and improving selected polders. Investments included civil works for (i) rehabilitating critical portions of polder embankments, including slope protection work, (ii) increasing embankment height in some stretches to improve resilience, (iii) repairing and upgrading drainage and flushing systems within polders, and (iv) improving operation and maintenance (O&M). At the appraisal stage, 17 polders across seven districts (Khulna, Bagerhat, Pirojpur, Patuakhali, Barguna, Jhalkathi, and Satkhira) were selected to be rehabilitated and improved with three works packages. The first works package covered four polders (32, 33, 35/1, and 35/3), the second works package covered six polders (39/2C, 40/2, 41/1, 43/2C, 47/2, and 48), and the third works package covered seven polders (14/1, 15, 16, 17/1, 17/2, 23, and 34/3).

b. **Subcomponent A2: Afforestation.** The objective of this component was to carry out afforestation activities for the resilience of embankments and the livelihoods of communities to provide protection from tidal flooding and storm surge.

12. Component B – Implementation of Social and Environmental Management Frameworks and Plans (original cost: US\$56.00 million; actual cost: US\$49.28 million)

a. **Subcomponent B1: Implementation of Social Action Plans.** This component was to support the consultation with and strengthening of polder stakeholders and beneficiaries by establishing water management organizations (WMOs)⁴ in all polders to determine the competing needs and uses for water resources, and to support the operation and maintenance (O&M) of hydraulic infrastructure.

b. **Subcomponent B2: Implementation of Social Management and Resettlement Policy Framework (SMRPF)**

⁴ WMOs refer to all types of organizations within a polder or scheme for participatory water management that includes Water Management Group (WMO) and Water Management Association (WMA). WMO is formed for each hydraulic boundary known as Water Management Unit (WMO) within a polder/scheme. There are several WMOs in a polder/scheme having WMO for each. An apex body of the WMOs is formed with representatives from each WMO called WMA for each polder/scheme. The WMOs and the WMA in a polder/scheme are known as WMOs.



and Resettlement Action Plans (RAPs). This subcomponent supported the implementation of the SMRPF and RAPs and financed land acquisition and the resettlement and rehabilitation of persons adversely affected by the project.

c. Subcomponent B3: Implementation of Environmental Management Framework and Environmental Management Plan (EMP). This component financed (i) the preparation of environmental impact assessments (EIAs) for all remaining polders, (ii) the implementation of the EMP and environmental mitigation and enhancement measures, and (iii) the establishment of an environmental monitoring system within the BWDB to enable it to track continuous improvement in the environmental performance of the polder system. Some of the items under the EMP were integrated into the civil works and included in the budget of Subcomponent A1.

13. Component C – Construction Supervision, Monitoring and Evaluation of the Project, and Coastal Zone Monitoring (original cost: US\$32.00 million; actual cost: US\$36.61 million)

a. Subcomponent C1: Detailed Design and Construction Supervision. This subcomponent covered consulting services for (i) surveys and designs of the remaining polders to be included in the project (other than the five for which detailed designs had already been completed) and (ii) construction supervision of the coastal embankment rehabilitation and improvement work.

b. Subcomponent C2: Third Party Monitoring and Evaluation of the Project. This subcomponent covered consulting services for continuously monitoring project activities and providing feedback to the GoB and the implementing agency on the project's performance, including supervising the implementation of the Governance and Accountability Action Plan, the EMP, and the RAPs.

c. Subcomponent C3: Long-Term Monitoring, Research, and Analysis of Bangladesh Coastal Zone. This subcomponent supported comprehensive monitoring and a morphological assessment of the Bangladesh Delta, as well as an extension of the current monitoring systems in coastal Bangladesh to generate data, information, and new knowledge for assessments of the effects of multiple drivers on the environment of the coastal zone to guide future design, rehabilitation, and improvement requirements.

14. Component D – Project Management, Technical Assistance, Training, and Strategic Studies (original cost: US\$21.00 million; actual cost: US\$10.38 million). This component supported the BWDB in implementing the project through **Subcomponent D1: Project Management Support and Audits**, whereby a fully functioning project management unit (PMU) was established and maintained and all necessary audit reports financed; **Subcomponent D2: Technical Assistance and Training**, whereby institutional capacity building, technical assistance, and training for the BWDB was provided; and **Subcomponent D3: Strategic Studies and future Project Preparation**, whereby resources were provided for needed strategic studies (including the continuous updating of the strategic polder assessment as well as all necessary preparatory studies for subsequent phases of the CEIP).

15. Component E – Contingent Emergency Response (original cost: US\$0 million IDA; actual cost: US\$0 million IDA). This component was included to allow the GoB to request the recategorization and reallocation of financing from other project components to partially cover emergency response and recovery costs.

B. SIGNIFICANT CHANGES DURING IMPLEMENTATION (IF APPLICABLE)

16. The project underwent three Level-2 restructurings: in January 2020, March 2022, and December 2023.



Revised PDOs and Outcome Targets

17. The PDOs were not revised, but the restructuring in January 2020 revised the end targets of two of the PDO-level indicators. The end target for the indicator “gross area protected” was decreased from 100,800 hectares to 66,012 hectares, and the end target for the indicator “direct project beneficiaries from increased resilience to climate change” was decreased from 760,000 to 724,202.

Revised Components

18. The January 2020 restructuring scaled down the scope of Component A from 17 to 10 polders, along with the corresponding targets for embankment upgrades and the replacement and repair of hydraulic structures. The December 2023 restructuring reduced the costs of Components A, B, C, and D.

Other Changes

19. **Extension of closing date.** The closing date was extended twice, resulting in a total extension of 36 months. In the January 2020 restructuring it was extended by 18 months, from December 31, 2020, to June 30, 2022. In the March 2022 restructuring, it was extended by 18 additional months, up to December 31, 2023.

20. **Changes in the results framework.** In addition to the changes to the PDO-level indicators, the January 2020 restructuring dropped, added, and revised various intermediate results indicators (IRIs) and their targets. Key changes included: a decrease in the end targets for the “length of upgraded embankment” IRI 1 from 623 to 408 kilometers, and for the “length of drainage channels upgraded” IRI 5 from 794 to 305 kilometers. The IRI 3 on upgrading regulators was dropped, and the IRIs on drainage structures and flushing inlets (IRI 2 and 4) were redefined, with lower end targets overall. End targets for a few IRIs were increased, including for the “area afforested” (from 300 to 600 hectares), the “number of water management associations” (from 4 to 10), and the “number of grievance redress committees” (from 17 to 36). Other changes and respective details are presented in Annex 1.

21. **Cancellation.** In the December 2023 restructuring, a US\$24.26 million IDA credit (SDR 18.223 million equivalent), was cancelled, revising the total project cost to US\$375.74 million.

Rationale for Changes and Their Implication on the Original Theory of Change

22. **The adjustment to the project scope during the January 2020 restructuring was because of cost overruns**, which were the result of (i) increased land acquisition costs (of 300 percent) mostly because of the new Acquisition and Requisition of Immovable Property Act 2017, which was passed during project implementation; (ii) increased value added tax and income tax;⁵ (iii) price escalation (increased unit costs, since the project was approved in June 2013); (iv) a change in the US\$/SDR exchange rate (resulting in a decrease from US\$375.00 million equivalent to US\$342.15 million equivalent as of November 7, 2019); and (v) the need to carry out emergency protection works during implementation (as described in the next paragraph).

23. **Severe riverbank erosion threatened the land, property, and livelihoods of local farmers and fishers, triggering the need for additional budget to carry out bank protective works.** Inland riverbank erosion, which can reach up to 150-200 meters in a single rainy season, necessitated the retirement of embankment sections prior to construction, and numerous sections required urgent rehabilitation. Without additional bank protective works, the land and property that were previously protected by these embankments would be at risk of being lost to erosion, impacting the lives and

⁵ A 2.0 percent increase in value added tax and 2.5 percent increase in income tax for works, and a 10 percent increase in income tax for consultancy services.



livelihoods of the local community, who are mostly impoverished farmers and fishers. Thus, the project was in need of additional inputs to carry out these protective works at the eroded sites.

24. The changes made to the PDO-level indicators, IRIs, and targets in the January 2020 restructuring aimed to reflect the revised scope of the project and enhance the measurement of project outcomes and results. The proposed new IRIs were meant to measure outputs that were already part of the project scope but were not reflected in the original results framework.

25. **The cancellation of** the US\$24.26 million (SDR 18.223 million) was because of (i) savings from works contracts (the final quantity and value of the works delivered was lower than the original contract's value and contingencies), and (ii) exchange rate fluctuations (exchange rates in some contracts were fixed at the time of signing leading to savings from the SDR/US\$ appreciation against the BDT).

26. **The extensions of the closing date** provided the Client with additional time to complete delayed activities (refer to Section III.B for details on the factors that contributed to the delays).

II. OUTCOME

A. RELEVANCE OF PDOs

Assessment of Relevance of PDOs and Rating

27. **The PDOs are highly relevant to the WBG's CPF for Bangladesh FY2023–27 (Report No. 181003-BD).** It is particularly relevant to the CPF's High-Level Outcome C: Enhanced climate and environmental resilience (CPF Objective 7). The project supported CPF Objective 7, which focuses on improved effectiveness of delta management to accelerate climate resilience building through activities such as the implementation of the Bangladesh Delta Plan 2100, building climate-resilient infrastructure and improved disaster risk management, and community-led adaptation and risk management. The number of people benefiting from enhanced resilience to riverbank erosion or inundation damage is one of the CPF's progress indicators that was supported by this project. The Project is also relevant to High-Level Outcome B: Improved socioeconomic inclusion, through enhancing economic opportunities for vulnerable groups, through improvement in agriculture resilience.

28. **The PDOs are also relevant to the visions and priorities of the GoB.** The project contributed to achieving the goals of Bangladesh's Eighth Five Year Plan FY2020–25 by strengthening community capacity, building resilient infrastructure, and providing technical assistance for O&M. Furthermore, it contributed to achieving the objectives of the Bangladesh Delta Plan 2100, which highlights the need to effectively reduce vulnerability to storm surges and build resilience to climate change and other delta challenges. The project is also consistent with Bangladesh's Nationally Determined Contribution, which was updated in August 2021. It identified CEIP-I as one of the key adaptation actions in mitigating some of the most significant impacts of cyclones and enhanced emergency response in the coastal region. Furthermore, the project is aligned with mitigation actions of the updated NDC such as afforestation activities with collaborative forest management and social forestry.

29. **The PDOs had an appropriate scope** given the considerable impacts of cyclone-induced storm surges on embankments, hydraulic structures, drainage channels, and cropping intensity in the selected polders. The downscaling of targets during implementation was the result of design and implementation constraints, rather than the reduced



relevance of the PDOs, as evidenced by the numerous cyclones⁶ that hit the coastal areas of Bangladesh during the project period.

B. ACHIEVEMENT OF PDOs (EFFICACY)

Assessment of Achievement of Each Objective/Outcome

30. **The achievement of the PDOs is evaluated before and after the January 2020 restructuring, which made changes to the project's indicators and targets.** The evaluation is structured around the three PDO outcome statements as shown in Table 1 using the original and revised/new indicators and other relevant results that can be directly attributed to the project and support the achievement of the project outcomes.

Outcome 1: Increased area protected in selected polders from tidal flooding and frequent storm surges, which are expected to worsen due to climate change (Table 2)

Table 2. Outcome 1 – Original and Revised Indicators, Results, and Achievement Rates

PDO-Level Indicator/IRI	Achieved	Target	% Achievement
Original: Gross area protected (1,000 x ha)	65.657	Original: 100.800 Revised: 66.012	Original: 65.1% Revised: 99.5%
Original: Direct project beneficiaries from increased resilience to climate change (number: 1,000 x person) of which female (percentage)	720.573 (50%)	Original: 760.000 (50%) Revised: 724.202 (50%)	Original: 94.8% Revised: 99.5%

Source: BWDB. CEIP-I project documents.

31. **The project protected selected coastal polders against the risk of flooding, storm surges, coastal erosion, and salinity intrusion.** It protected a gross area of 65,657 hectares across 10 polders, achieving 65 percent of the original target of 100,800 hectares (partly achieved), and almost fully meeting the revised target of 66,012 hectares. Additionally, the project provided increased resilience to climate change for 720,573 individuals, almost fully achieving the revised target of 724,202 individuals and the original target of 760,000.

32. **This positive achievement was the result of the upgrading of embankments, afforestation, and the construction of riverbank and slope protective works.** The embankments were upgraded to resilient standards with top width, side slopes, and free board considering historical storm surge levels and climate change impacts. Originally, there were 623 kilometers of embankment planned. However, when the number of polders was decreased from 17 to 10, the target was decreased proportionately. The project almost achieved (405.67 kilometers) the revised target of 408 kilometers and fell short of achieving the original target of 623 kilometers. As an initiative of a nature-based solution concept embedded into the project design for climate resilience, the project afforested a total area of 700 hectares, exceeding the revised target of 600 hectares and the original target of 300 hectares. The project also exceeded the results framework targets for slope protective works (28 kilometers), completing 29.09 kilometers, and bank protective works (9.37 kilometers), completing 9.85 kilometers.

33. **The project improved the understanding of the coastal zone substantially and generated very useful results for the next generation of coastal programs.** The analytics component, the long-term monitoring, research, and analysis of the coastal zone activities completed all numerical modeling of the long-term physical processes and other relevant phenomena (such as subsidence, climate change, river morphology) (scale: macro, meso, and micro) for the current

⁶ Cyclone Roanu (2016), Cyclone Mora (2017), Cyclone Fani (2019), Cyclone Bulbul (2019), Cyclone Amphan (2020), Cyclone Yaas (2021), Cyclone Jawad (2021), Cyclone Sitrang (2022), Cyclone Mocha (2023), Cyclone Hameon (2023), and Cyclone Midhili (2023).



and future scenarios. The studies generated a wealth of knowledge and understanding of the coastal system, developed practical tools for beneficiaries to better manage the system (for example, bank erosion forecasting tools), and provided training to beneficiaries on how to use the knowledge and tools for next generation coastal interventions. An investment plan that describes a phased polder improvement roadmap for coastal Bangladesh and the underlying data collection, including socioeconomic data, has also been completed, fulfilling the results framework target dedicated to these analytics. Based on these analytics, the project prepared the feasibility study and detailed design for the next phase of the coastal resilience project.

34. The project grievance redress mechanism (GRM) was accessible to the communities and affected persons. Grievance redress committees (GRCs) were formed and operated in each Union Parishad area. A total of 36 GRCs were formed in the project areas of the 10 polders, exceeding the original target of 17, with representatives from the BWDB, the Union Parishad, local educational institutions, project-affected persons, women's groups, and the RAP team. A GRM leaflet was distributed to project-affected persons and their communities in the polder areas with contact details of the GRM focal points at the local level, the office of the Project Director, and the Bank's grievance redress service.

35. The project also completed the detailed design, including safeguards studies, for the seven polders that were cut from the project in the January 2020 restructuring. These seven polders were moved to a potential second phase of the project together with additional polders for which feasibility study and detailed design were also done. The source of financing for the second phase has not yet been determined.

Outcome 2: Improved agricultural production by reducing saline water intrusion in selected polders (Table 3)

Table 3. Outcome 2 – Original and Revised Indicators, Results, and Achievement Rates

PDO-Level Indicator/IRI	Achieved	Target	% Achievement
Original: Increase cropping intensity (%)	192	Original: 180 Revised: No change	106.7%

Source: BWDB. CEIP-I project documents.

36. The project significantly enhanced agricultural efficiency and community resilience by enabling nearly two crop cycles annually. The cropping intensity at the project sites increased from a baseline of 140 percent to 192 percent,⁷ exceeding the results framework target of 180 percent. This marks a pivotal enhancement of agricultural practices. This achievement translates into several tangible benefits for the community, including optimized land use, increased food production, higher incomes for farmers, improved food security, and greater crop diversification. Additionally, a comparison of the cultivation of a high-value crop like watermelon between the CEIP-1 polders under the first works package and non-CEIP-I polders highlights the significant impact of the CEIP-I agricultural interventions on cropping patterns. In the 2022-23 reporting period, watermelon cultivation in the first works package polders reached almost twice the cropped area compared to non-CEIP-I polders, whereas in 2015-16, the watermelon cropped area was negligible in the polders under the first works package.⁸

37. This achievement was the result of the construction and repair of drainage and flushing structures, the

⁷ Cropping intensity (the number of crops a farmer grows in a given agricultural year on the same field) is calculated by taking the total gross cropped area (single, double, and triple crops) and dividing it by the total cultivable land in a cropping season for the 10 selected polders, expressed in %.

⁸ In 2022-23 Watermelon cultivation in the first works package polders reached 9 percent of the total cropped area, compared to only 5 percent in non-CEIP-I polders. In 2015-16, it was 3% for first works package polders and 1% in non-CEIP-I polders. See GoB Bangladesh Water Development Board. 2023. *CEIP-I Final Impact Evaluation Report*.



upgrading of drainage channels, and the formation of water management associations (WMAs). As a result of the project, 89 drainage structures and 85 flushing structures were built, and 46 existing flushing inlets and 8 existing drainage structures were repaired, all either achieving or exceeding the results framework end targets. For the upgrading of drainage channels, originally 794 kilometers of channels were planned; however, as a result of the January 2020 project restructuring, the drainage channel target was decreased proportionate to the reduction of the project scope from 17 to 10 polders. The project achieved (305.85 km) the revised target of 305 kilometers but fell short of achieving the original target of 794 kilometers. The project also made significant progress in community mobilization. With the formation of 10 WMAs, the communities are overseeing the overall polder O&M as well as the minimal upkeep needed during cyclones to ensure overall sustainability (such as closing the gates to avoid tidal flooding).

Outcome 3: Improved capacity of the Government of Bangladesh to respond promptly and effectively to an eligible crisis or emergency

38. **This element of the PDO statement refers to the Contingency Emergency Response Component (CERC), which is only triggered in the event of the declaration of an emergency.** The achievement of Outcome 3 is not rated since the CERC was not triggered. It is worth noting that recent Bank guidance for projects that include CERCs advises that the PDO statement need not include a reference to the CERC (since it might not be triggered) or should be followed by “if applicable” so that projects are not held accountable for meeting this objective unless the CERC is triggered.

Justification of Overall Efficacy Rating

39. **The overall efficacy of the project is rated Substantial both before and after the January 2020 restructuring** based on the levels of achievement of the first two outcome statements (the third outcome was not rated (see paragraph 38) using the original and revised targets as described in Table 4. Although the Outcome 1 rating before restructuring is Modest because the gross area protected was partly achieved, the operation almost fully achieved its objective of increasing resilience to climate change for the original target population. This achievement carries significant weight when evaluating the project’s efficacy before the January 2020 restructuring, since it directly contributed to the well-being and adaptability of the communities in the face of climate change, with increased nature-based solutions. Furthermore, the rating accounts for the value added through Outcome 2 beyond the achievement of the envisioned scale at appraisal, including the criticality of investments for water management inside the polder, with hydraulic structures, drainage channels, and WMOs, as demonstrated by the performance of investments during multiple cyclone events during and after the project period. This rating also considers the large evidence base compiled through the improved coastal monitoring system, and modeling exercises and tools created, to allow for the development of risk-informed investment plans for polders to withstand extreme climate events.

Table 4. Explanation of the Efficacy Rating Before and After the January 2020 Restructuring

Outcome	Efficacy Rating	
	Before	After
Outcome 1: Increased area protected in selected polders from tidal flooding and frequent storm surges, which are expected to worsen due to climate change	Modest	Substantial
Outcome 2: Improved agricultural production by reducing saline water intrusion in selected polders	High	High
Outcome 3: Improved capacity of the Government of Bangladesh to respond promptly and effectively to an eligible crisis or emergency	Not Rated	Not Rated
Overall Rating	Substantial	Substantial

Source: BWDB. CEIP-I project documents.



C. EFFICIENCY

Assessment of Efficiency and Rating

40. **The project's overall efficiency is rated as Modest, despite its high economic efficiency, due to its modest implementation efficiency.**

Economic Efficiency

41. **An economic analysis was conducted to assess the rate of return of the capital investments of two components—A. Rehabilitation and Improvement of Polders and B. Implementation of Social and Environmental Management Frameworks and Plans.** The total cost of Components A and B was US\$264.80 million, or 85 percent of the total project cost of US\$311.84 million. See Annex 4 for details.

42. **The expected values of the economic indicator matrix of the economic internal rate of return (EIRR), net present value (NPV), and benefit-cost ratio (BCR) are in the high range.** The EIRR for the whole project was 25.7 percent. The NPV was in the range of US\$372.80 to 1,177.50 million and the BCR was in the range of 2.9 to 5.1 for discount rates of 12 percent and 6 percent, respectively. The NPV at appraisal was calculated using a 10 percent discount rate. Thus, it is not surprising that the NPV at appraisal was within the range calculated in this analysis. Despite differences in coverage and methodology, the economic indicators at project completion are similar to those at appraisal.

Implementation Efficiency

43. **The project's implementation efficiency was moderate, as evidenced by the completion of only 10 out of 17 planned polders, cost overruns, and a 36-month delay.** There were delays in completion of the contracts for polder rehabilitation attributed to implementation delays, and procurement and contract management issues, which were exacerbated by the COVID-19 pandemic. In areas vulnerable to recurring disasters, there were delays of up to 131 percent for the first works package (47 months), and up to 83 percent for the second works package (35 months). This increased the vulnerability to disaster risk of the population, agricultural land, and assets for a longer period than originally intended. For example, in some areas where the installation of gates to repair sluices or the construction of structures was delayed, the target beneficiary farmers were deprived of irrigation facilities for an entire year. Despite these delays and issues, there are inherent challenges associated with the timely completion of large infrastructure works, such as the rehabilitation of embankments, in remote and climate vulnerable areas, as outlined in Section III. Key Factors that Affected Implementation and Outcome.

44. **The project had an opportunity to improve its expenditure projections based on the actual scope of work to ensure that the GoB could fully utilize concessional funding from the IDA.** Following the disbursement grace period, which included a 15-day extension for documentation purposes on an exceptional basis, the project had an undocumented designated account balance of BDT 730,143,477.87, equivalent to US\$6.24 million, along with an additional US\$6.45 million as an undisbursed balance. As a result of not cancelling these balances, the GoB will forfeit a total of US\$12.69 million in concessional IDA funding. This situation highlights the inability of CEIP-I to accurately forecast actual expenditures based on the final scope, since the cancellation that was done in December 2023 could have included the US\$12.69 million, suggesting a lack of foresight and efficient FM on the part of the project.

45. **Despite cost overruns and additional resource requirements for consultants because of project extensions, the project demonstrated some operational efficiency by completing the rehabilitation of 10 polders mostly within**



cost estimates and maintaining lower project management costs at closing compared to the appraisal estimates. Between 2013 to 2019, the project was subject to increased costs for land acquisition, value added tax, and income tax. Furthermore, in line with the extensions of the project closing date, additional inputs and resources were required for detailed design construction supervision and project management support (DDCS & PMS) (a 41.42 percent increase) and monitoring and evaluation (M&E) consultants (a 48.13 percent increase) to supervise the works and associated M&E activities under Component C. Conversely, the project management costs under Component D at closing were 50 percent of the cost determined at appraisal suggesting operational efficiency. Furthermore, the rehabilitation of 10 polders was mostly completed within the cost estimates, except for the adjustment in contract prices made to accommodate the change in tax legislation.

D. JUSTIFICATION OF OVERALL OUTCOME RATING

46. **The overall outcome is rated Moderately Satisfactory.** The rating is based on the project's High relevance, Substantial efficacy before the January 2020 restructuring, Substantial efficacy after the restructuring using the revised indicator targets, and Modest efficiency (see Table 5).

Table 5. Calculation of Outcome Rating

	Original Objectives and Targets without Restructuring	Revised Objectives and Targets with Restructuring
Relevance of PDOs	High	
Efficacy (PDOs)	Substantial	Substantial
Outcome 1	Modest	Substantial
Outcome 2	High	High
Outcome 3	Not Rated	Not Rated
Efficiency	Modest	
Outcome ratings	Moderately Satisfactory	Moderately Satisfactory
Numerical value of the outcome ratings ^(a)	4	4
Disbursement (US\$ million)	203.76 ^(b)	126.84 ^(c)
Share of disbursement ^(d)	0.6163 (or 61.63%)	0.3837 (or 38.37%)
Weighted value of the outcome rating	2.4653	1.5347
Final outcome rating ^(e)	Moderately Satisfactory (2.4653 + 1.5347 = 4.00)	

Source: The World Bank. Original table developed for this report.

Note: (a) Ratings are: 1. Highly Unsatisfactory; 2. Unsatisfactory; 3. Moderately Unsatisfactory; 4. Moderately Satisfactory; 5. Satisfactory; 6. Highly Satisfactory.

(b) During January 2020, total disbursement was US\$203.76 million—IDA-52800 = US\$178.77 million and TF-14713 = US\$24.99 million.

(c) As of May 15, 2024, total disbursement was US\$336.84 million—IDA-52800 = US\$311.84 million and TF-14713 = US\$25.00 million. However, the project had an outstanding balance of BDT 730,143,477.87, equivalent to US\$6.24 million, resulting in an actual disbursement for expenditures incurred of US\$330.60 million. Deducting the January 2020 disbursement of US\$203.76 million from that equals US\$126.84 million.

(d) After deducting outstanding balance of US\$ 6.24 million from the disbursed amount, total disbursement with expenditure would be US\$330.60 million, which would be the denominator.

(e) See *Bank Guidance on ICR of IPF Operations (OPS5.03-GUID.156)*, Appendix H. Deriving the Overall Outcome Rating.



E. OTHER OUTCOMES AND IMPACTS

Gender

47. **At appraisal, the project recognized that the increased risk of storm surge and tidal flooding from climate change disproportionately affected the livelihoods of women in coastal polders, since their daily activities were closely related to natural resources and therefore were at high risk from climate variability.** However, the project design did not include explicit indicators to address this gap. The results framework included a generic gender-disaggregated sub indicator that only monitored the proportion of female direct beneficiaries. At closing, the Client reported that 50 percent of the total direct beneficiaries of the project were female. The project considered a generic 50 percent female beneficiary rate, primarily because of data quality issues and the lack of a clearly defined methodology during project appraisal for this indicator and the target. As a result, this indicator cannot effectively measure the project's efforts to address the gender gap.

48. **The project design included activities aimed at addressing this gap.** Women's participation in both the general membership and executive committees of water management groups (WMGs) and WMAs meets the required threshold of 30 percent female representation set by the Bangladesh Participatory Water Management Rules 2014. To date, 141 WMGs have been formed, with women constituting 36 percent (25,878) of the 72,329 members. Additionally, 561 women hold positions on the executive committees of the 10 registered WMAs, achieving a 33 percent female participation rate. The afforestation scheme guaranteed women participation as members of social forestry groups (a total of 2,351 female beneficiaries out of 8,278). Local women were employed as watchers as part of the afforestation program, where they took care of the planted seedlings while also earning a livelihood. Under the first works package, a total of 291 female-headed households were impacted because of resettlement, and under the second works package, there were 478 such households. The women household heads received special relocation and resettlement benefits and were also provided with a grant.

Institutional Strengthening

49. **The project has been instrumental in enhancing the institutional capabilities of the BWDB.** As a result of the project, 1,200 person-days of training were provided to BWDB officials, fully achieving the results framework target. Another project result of institutional strengthening not measured by the results framework was the long-term training of three BWDB officials, who received their Master of Science degrees from the IHE Delft Institute for Water Education in the Netherlands. This helped to increase the BWDB's theoretical understanding of, and practical skills in climate change resilience, polder management, and long-term investment planning.

Poverty Reduction and Shared Prosperity

50. **By enhancing the disaster resilience of poor households in the targeted coastal communities, the project contributed to poverty reduction.** Between 2010 and 2016, Bangladesh's poverty reduction pace slowed despite accelerated economic growth, with consumption growth of the bottom 40 percent trailing the overall population. However, the project interventions better protected poor communities' economic opportunities, livelihoods, and well-being through safer embankments that can withstand storm surges. Embankment rehabilitation generated short-term employment, with 1,147,064 workdays created for the 405.67 kilometers completed. Rehabilitated embankments increased adjacent agricultural land value by 100-200 percent, alleviating poverty and benefiting disaster-affected



people. Social afforestation initiatives aimed to reduce poverty by allocating plantation sales proceeds to support 8,278 vulnerable beneficiaries.

Other Unintended Outcomes and Impacts

51. **The project led to a substantial net reduction in greenhouse gas emissions, with carbon sequestration through afforestation activities under Component A.** The project's plantation activities across a total of 700 hectares resulted in the sequestration of about 117,814 metric tons of CO₂ equivalent. The activities also increased the green coverage of the coastal area, thereby enhancing the aesthetics along the roads, and supporting the mitigation of heat waves, air pollution, and related public health problems.

52. **The embankments rehabilitated as part of CEIP-I also serve as roads. Prior to polder rehabilitation, in more remote areas of the polders, it was difficult for farmers to travel and access markets.** Now, as polders have been rehabilitated, business connections and communication networks have redeveloped, and people use the embankments to carry their crops to central markets for sale. This has helped boost the local economic impacts of the project.

53. **The project supported inclusive community participation and livelihoods adaptation for sustainable resilience.** It formed 141 WMGs with 72,329 members and 10 WMAs with 1,692 members, all of which are now registered with the BWDB Principal Extension Office. These WMAs and WMGs are responsible for the polders' O&M under a memorandum of understanding signed with the BWDB O&M divisions for all 10 polders. Furthermore, the project supported the capacity building of these groups with training activities delivered by nongovernmental organizations it engaged. Additionally, the project formed 142 social forestry groups with 8,278 beneficiaries for afforestation, which ensures a structured benefit-sharing scheme from the sales proceeds after the customary rotation period.

III. KEY FACTORS THAT AFFECTED IMPLEMENTATION AND OUTCOME

A. KEY FACTORS DURING PREPARATION

54. **The project set the PDOs with a mixed level of ambition.** The first two objectives, focusing on increasing the area protected from tidal flooding and frequent storm surges and improving agricultural production by reducing saline water intrusion, are at the right level of ambition given the existing 6,000 kilometers of embankments with 139 polders. The magnitude of such a project is enormous, and the adoption of a multi-phased approach, with CEIP-I being the first phase, reflects a realistic and strategic approach to tackling this challenge. A single investment loan was deemed most suitable to achieve these objectives in a complex and changing environment and to pilot innovative concepts, with the potential for a series of projects to be designed for other exposed coastal regions of Bangladesh based on CEIP-I's success. However, the third objective, aiming to improve the GoB's capacity to respond promptly and effectively to an eligible crisis or emergency, does not reflect the same level of ambition and appears to have no focused project activities to achieve it.

55. **The project design effectively addressed the GoB's request but could have better aligned its scope and structure with the PDO outcomes, given the extensions and reduced activities.** The project design responded well to the GoB's request for a long-term disaster management program. The Emergency Cyclone Recovery and Restoration Project (P111272) supported the planning and design of CEIP-I, which included protection against a 25-year return period maximum surge height, with an additional buffer for climate change impacts. Considering that the project had



to be extended two times and some activities were reduced in scale for various reasons, the project's scope, particularly related to Outcome 1 supported by Components A, B, C, and D, had some room for improvement. Additionally, the project component structure could have been organized in closer alignment with the PDO outcome statements (instead of, for example, Component A financing activities that supported both Outcomes 1 and 2) to help in evaluating the achievement of each outcome.

56. **The risks were not properly assessed, and adequate mitigation measures were not fully identified at project appraisal.** The capacity risk of the implementing agency, though identified as Substantial, was not properly assessed, and adequate measures were not fully identified to address the capacity issue in implementing large value works contracts on time, which should have been a driving risk for the project. Furthermore, the project design by the Bank was not aligned with the GoB's Development Project Proforma (DPP), despite the design risk being identified as Moderate. For example, the end target of physical interventions and the allocated costs for different line items differed between the DPP and the PAD.

57. **Despite having key procurement and technical documents ready at effectiveness, the project took nearly two years to establish all implementation arrangements, which was not anticipated or addressed during appraisal.** At effectiveness, the project had in place the required detailed design, RAP, EIA, and bid documents for the first works package, which included four polders (Polders 32, 33, 35/1, and 35/3). Terms of reference for the associated consultancy service contracts—DDCS&PMS consultancy services, M&E consultancy services, procurement panel, independent panel of experts, and other key individual consultants for the PMU—were also prepared. However, it took almost two years for all implementation arrangements to be put in place, for the reasons discussed below. Furthermore, the project did not anticipate, identify, and properly address these delays at the time of appraisal.

B. KEY FACTORS DURING IMPLEMENTATION

Factors subject to the Client's control

58. **Delays in initial contracting for works and consultancies significantly delayed implementation after effectiveness.** For example, the DDCS&PMS consultant was only mobilized in January 2015, over one year after effectiveness because of the slow government approval process for large-value service contracts. The first works contract package was signed on November 1, 2015, almost two years after effectiveness because of a change in the procurement process. The task team recommended changing to post-qualification bidding for this package from pre-qualification to enhance competition and attract more favorable bids. However, both these works and service packages were ready for bidding prior to effectiveness. The second works package faced substantial delays in the issuance of the notification of award, the signing of the contract, and the contractor's mobilization because of the BWDB's slow process in preparing the EIA reports, getting approval from the competent authority, and lack of adequate internal fiduciary controls. For the long-term monitoring, research, and analysis of the coastal zone activities, the signing of the contract took almost two and a half years after the opening of the financial proposal because the DPP allocation was inadequate and not in line with the PAD.

59. **Limited contractor capacity, design modifications, and supplementary works posed significant obstacles to efficient project execution, exacerbated by the BWDB's internal contract management challenges.** Mobilization and the physical progress of both works packages were initially very slow. There were also delays in providing designs and drawings (such as the detailed design for the flushing sluice in Polder 43/2C and the termination of slope protective works in Polder 39/2C). The embankment construction work for Polders 39/2C and 48 was substantially delayed because of insufficient nearby borrow material (soil). The mid-term review (MTR) mission found that the environmental,



health, and safety management measures were not being carried out fully as per the agreed EMP for each polder. There was also a delay in the preparation of environmental action plans for the polders under construction. These were exacerbated by several contract management issues within the BWDB that hindered project progress. For example, the BWDB had a task force for overseeing the quality of cement concrete blocks used in protective works, but there was no standard procedure to ensure timely review. At the World Bank's suggestion, a standing order was issued that required the task force to inspect cement concrete block production weekly at a designated day and time, streamlining the process.

60. The CEIP-I designs did not fully anticipate the extensive erosion caused by the morphological changes of the rivers, necessitating supplementary bank protection measures to ensure the polders' long-term stability and resilience. The CEIP-I designs, prepared in 2012, were based on an assessment of the riverbank erosion at that time. However, frequent morphological changes in river alignment caused severe erosion at new sites not covered by the original contract. To ensure the safety of the polders and people, additional bank protection work was deemed necessary. After a lengthy review process, the existing contractors declined to carry out the additional work because of disagreements on construction rates. With limited time remaining, the PMU revised the design approach and executed emergency protective works through the BWDB's O&M division as Deposit Work in the final six months of the project.

61. A combination of factors created significant delays in the land acquisition process. The process itself is inherently slow and multilayered, typically taking 12 to 15 months from initial surveys to compensation delivery. This timeline is further extended by complexities related to land titling, which arise from incomplete and faulty ownership records. However, the PMU also contributed to the delays by submitting inconsistent LAPs that deviated from the approved RAP. This discrepancy triggered the withdrawal of the initial LAP and submission of a revised version, adding to the overall delay. Furthermore, even after the LAP was approved, shifts in the embankment alignment caused by river erosion necessitated changes to the LAP, further prolonging the entire process. These compounding factors, both inherent to the land acquisition process and stemming from the PMU's actions, resulted in substantial delays in acquiring the necessary land for the project.

Factors subject to the World Bank's control

62. The project's progress was impeded by the delayed implementation of the MTR mission's plans and poorly timed restructurings. The MTR mission in January 2017 thoroughly analyzed the implementation delays and cost overruns experienced during the first half of the project and developed plans to remove bottlenecks and better measure progress. However, these plans were not fully and timely executed by the World Bank. With the official revision of the Results Framework taking place three years after the MTR. The Bank considered providing additional financing to cover cost overruns, contingent upon improvements in project implementation and safeguards compliance, but the Client did not meet these conditions. The timing of the three restructurings affected their effectiveness, with the most fundamental restructuring (including to the results framework) occurring three years after the MTR in January 2020, missing a critical opportunity to address setbacks earlier. The second restructuring took four months of review, while the last restructuring, which changed component costs and cancelled IDA funds, took place only 17 days before closing.

Factors outside the control of the Client and the World Bank

63. Strikes and blockades: Numerous hartals (strikes) and blockades took place during the project period, with the most occurring during November 2013 to January 2014, January 2018 to January 2019, and December 2023 (just before the project close). This hampered project implementation, supervision, monitoring, and financial and operational delivery.



64. **Natural disasters, and extreme weather:** The work plan targets were negatively impacted by damage from Cyclone Yaas in May 2021. In addition to the cyclone, there had been unprecedented rainfall and associated flooding at the project sites. These events not only hindered project implementation but also affected critical areas such as contractors' sites and marketplaces.

65. **COVID-19 pandemic.** The pandemic posed a major challenge since it brought project implementation to a standstill, with nationwide lockdowns, concerns over health risks to staff, and the follow-on effect of material shortages and supply chain issues. Several staff and consultants from the implementing agency and line ministry were personally affected by COVID-19, both at the headquarters and district levels.

IV. BANK PERFORMANCE, COMPLIANCE ISSUES, AND RISK TO DEVELOPMENT OUTCOME

A. QUALITY OF MONITORING AND EVALUATION (M&E)

M&E Design

66. **The project's M&E system had some limitations in assessing the project's progress and impact, with a few indicator definitions lacking clarity; however, the project had a sufficient set of indicators overall to track its advancement and examine the connections within the results chain.** PDO-level Indicator 3 "increase cropping intensity," had a baseline of 140 percent and an end target of 180 percent, indicating an expected 29 percent increase over baseline at completion. For IRIs, the difference between upgradation, replacement, and regulators was not clearly defined. The substantial cost of slope and bank protective works was not reflected in the results framework, which also did not adequately measure the project's intended impact on gender gaps. Despite the potential, CEIP-I did not include climate indicators, although they were not a corporate requirement during preparation.

67. **The results framework was revised in the January 2020 restructuring, partially addressing the design shortcomings. Missing IRIs were added to reflect the appropriate levels of measurement.**

M&E Implementation

68. **Data collection for the assessment of the PDOs, intermediate outcomes, and component activities was carried out during implementation through a third-party M&E consultancy firm.** The firm was responsible for overall monitoring and supervision of the implementation of various components, including subproject-level environmental management plans. M&E consultants were only mobilized in October 2015, almost two years after effectiveness.

69. **The collection of baseline data for the CEIP-I polders was adjusted to accommodate project delays and included comparison data from non-CEIP polders and surveys of both the general population and project-affected households.** Baseline data for the three works packages were collected in 2016, but because of substantial delays in the second works package, its baseline data were updated with 2017 agricultural data in 2018. For comparison, data were also collected from non-CEIP polders. Surveys were conducted on two groups: the general population (entire polder) and project-affected households that would lose their land, homes, or businesses because of the embankment works.



70. **The PMU consistently delivered timely and comprehensive M&E reports, established a web-based monitoring system, and provided a thorough analysis of the project's impact using relevant indicators.** The PMU's M&E team prepared and submitted comprehensive quarterly and annual progress reports to the World Bank consistently within 45 days of the end of the quarter. It established a web-based project monitoring information system for the project, piloting a partial beta version in November 2016, updating it in September 2017, and making an improved version available to users in December 2017. The PMU's *Final Impact Evaluation Report* provided a thorough analysis of the M&E data, going beyond the indicators outlined in the results framework to include more relevant indicators such as income, expenditure, poverty incidence, and the effectiveness of embankment utilization.

M&E Utilization

71. **The PMU communicated monitoring data to the various stakeholders, which were utilized to inform project management and decision-making by the Project Steering Committee.** Access to real-time information through the project monitoring information system helped the PMU identify and implement corrective measures to address the delays and noncompliance. In addition, the PMU publicly disclosed the project's monitoring information, including the M&E reports, periodic progress reports, safeguards reports, and financial status through the project website.⁹

Justification of Overall Rating of Quality of M&E

72. **The overall M&E quality is rated Substantial.** There were moderate shortcomings in the M&E design. The M&E system as implemented and utilized was generally sufficient to assess the achievement of the objectives and test the links in the results chain. Despite hurdles, the M&E data were utilized in an effective manner to track progress, monitor project activities, and record results against the PDOs.

B. ENVIRONMENTAL, SOCIAL, AND FIDUCIARY COMPLIANCE

73. **The project implemented comprehensive safeguards compliance with a framework approach** followed by site-specific safeguards assessments for the selected polders, complemented by robust community consultations, annual environmental clearances, continuous monitoring, and targeted environmental enhancement activities.

74. **Environmental: Overall, the environmental safeguards performance of the project was satisfactory, with a minor setback during the onset of construction.** The project was classified as a safeguards category "A" and triggered environmental safeguards policies for environmental assessment (operational policy (OP)/Bank procedure (BP) 4.01), Natural Habitats (OP/BP 4.04), Forests (OP/BP 4.36), and Physical Cultural Resources (OP/BP 4.11). The BWDB developed an environmental management framework for the project. It also prepared polder-specific EIAs for the original 17 polders and recommended polder-specific EMPs with polder-specific consultations with communities and stakeholders. An independent panel of experts with international experience reviewed all EIAs. The BWDB also disclosed the EIAs and environmental management framework on the project website and conducted a national-level consultation to finalize the documents. Environmental clearance certificates for the first two works packages were issued by the Department of Environment and renewed annually. Adequate EMP implementation costs were included in the Bill of Quantity items that were spent for implementation of the EMP-suggested mitigation measures. Contractors for the first and second works packages prepared contractor environmental and social management plans (C-ESMP), which were reviewed and cleared by the World Bank and updated based on the site requirements as and when needed. The contractors of both the first and second works packages conducted robust monthly environmental

⁹ <http://ceip-bwdb.gov.bd/front.html>.



training programs. In addition to the monthly and quarterly progress reports on environmental safeguards, the project prepared semiannual environmental monitoring reports and annual environmental audit reports. The project undertook several environmental benefit enhancement activities, including constructing fish conservation sanctuaries, studying fish catch assessment, conducting net pen culture, raising awareness on aquatic biodiversity conservation, and promoting social afforestation in the polder areas. At closing, the project prepared an environmental action plan based on the environmental and social management plan requirements for the operations phase.

75. Social: The project successfully managed social safeguards, including involuntary resettlement, grievance redress, and labor influx monitoring, while making significant progress in land acquisition and compensation payments. The project triggered OP/BP 4.12 on Involuntary Resettlement. Since none of the residents or affected persons within the polders selected by CEIP-I were from tribal communities with characteristics similar to Indigenous Peoples per OP/BP 4.10 on Indigenous Peoples, this policy was not triggered for CEIP-I. During appraisal, the project prepared an SMRPF, which included a resettlement policy framework and a social inclusion/gender framework, and RAPs for the first works package. The SMRPF guided the preparation of the RAPs for the remaining works packages and the updating of the RAPs for the first works package, which the BWDB prepared, obtained agreement from the Bank, disclosed locally, and implemented. The project established 36 GRCs across the first two packages, which received 269 complaints by December 31, 2023, and successfully resolved all cases through initial assessments or formal GRC procedures. Furthermore, the project provided monthly reports on labor influx risks, requirements, and implications for the works packages with a checklist for tracking labor influx-related issues.

76. By project close, the project had made significant progress on land acquisition and compensation payments and no construction was carried out until the process was complete. As of April 2024, land acquisition payment progress was 92.4 percent, with the remaining payments attributed to the incomplete embankment works, small value of compensation (even as little as BDT 345/US\$4.00), and lack of ownership decision on scrutiny of the record of rights for reasons such as the unavailability of mauza maps (cadaster maps), khatians (landholding records for the payment of land tax), transfer/succession documents, or possession records, as well as disputed claims with objections and parcels of land under sub-judicial cases. The BWDB has developed a system to monitor household-level compensation payment status and a time-bound action plan to address case-specific land acquisition payment bottlenecks. The Key Focal Point for the project, along with the executive engineers of the concerned BWDB O&M divisions, will track, catalyze, and coordinate with the Deputy Commissioner offices of concerned districts to complete payments to the remaining affected landowners and report to the Bank on a monthly basis. Resettlement assistance for involuntarily displaced non-titled persons was fully completed prior to the commencement of works for the respective sections.

77. Procurement: The BWDB fully complied with the World Bank's procurement guidelines and procedures. As part of the preparation process, a procurement capacity assessment was conducted of the BWDB. Various risk mitigation measures were proposed and implemented as planned, including but not limited to, establishing a Procurement Panel made up of two international consultants and one national consultant as well as conducting training on World Bank procurement procedures, international competitive bidding procurement, and fraud and corruption issues. The PMU had previous experience with World Bank procurement guidelines and procedures, resulting in overall good procurement performance. The procurement plans were updated and published in a timely manner and no procurement commenced without an approved procurement plan containing the activity.

78. Financial management: The BWDB fully complied with the World Bank's FM guidelines and procedures, with minor setbacks. The BWDB had prior experience in implementing Bank-financed programs and had reasonable internal controls and accounting system in place for the accounting and reporting of expenditures. The following arrangements governed the project financial arrangements: (i) an FM specialist, with terms of reference that were agreed with the



Bank, was recruited after some initial delays but maintained for the duration of the project, although the project had one dedicated Deputy Director (Accounts) and two dedicated accountants to support FM activities throughout the duration of the project; (ii) two separate bank accounts were opened for implementing the project; (iii) all payments were made directly by the Project Director from the PMU; (iv) the accounting system of the BWDB was used for accounting and reporting of the project expenditures; and (v) an annual external audit of the project was carried out by the Comptroller & Auditor General of Bangladesh, through its Foreign Aided Project Directorate (FAPAD), and audit reports were submitted to the World Bank in a timely manner. Performance audits were carried out by an independent private auditor on the basis of terms of reference and a selection process acceptable to the Bank. However, there has been an overall inordinate delay in resolving audit observations. The project has done well in settling all audit observations barring three raised by FAPAD as of FY2022. The outstanding observations are related to land acquisition; the FAPAD Director General is working separately on them because the issues are beyond the project's control. For the FY2023 external audit report, eight observations, seven serious financial irregularities and one non-serious financial irregularity were raised by the auditors. The project has sent responses to all the observations to FAPAD for resolution. Despite initial delays in completing internal (FY2013 to FY2016) and external audit reports, they were later conducted regularly.

C. BANK PERFORMANCE

Quality at Entry

79. **The World Bank prepared and approved a project that was responsive to the critical needs of the Client and its long-term objective of coastal resilience, but with minor weaknesses at entry.** The design was based on lessons learned from the Coastal Embankment Rehabilitation Project and the Emergency 2007 Cyclone Recovery and Restoration Project (ECRRP) and had a strong concept. The World Bank was prompt in mobilizing a multisectoral team of technical and operational staff and consultants to support the preparation of the project design, and adequately fulfilled its fiduciary role by carrying out procurement assessments and supporting the development and disclosure of necessary safeguards documents before appraisal. However, shortcomings at entry in the project design included weaknesses in the M&E design, insufficient attention to how to minimize identified gender gaps, and the underestimation of some risks related to the limited capacity of the BWDB in terms of contract management and land acquisition.

Quality of Supervision

80. **The World Bank team carried out its implementation support and supervision responsibilities satisfactorily, with minor setbacks.** The team was adequately staffed with the necessary expertise and was composed of mostly in-country operational staff, some international staff, and seasoned technical consultants who provided ample support to the Client. In-person, virtual, and hybrid missions took place approximately every six months as per travel guidelines (during the COVID-19 pandemic), along with 38 technical field missions, various reviews, and a number of high-level discussions with the BWDB and the Ministry of Water Resources. The Bank team also provided hands-on procurement, FM, and social and environmental safeguards training to government officials.

81. **The missions produced Aide Memoires that detailed project progress, implementation issues, and action plans, and 20 implementation status and results reports (ISRs) that reflected these while highlighting key issues for the Bank's management's attention.** All project documentation, including ratings, clearly reflected the challenges and progress of the PMU. Delays in decision-making and implementation were clearly outlined in Management Letters, Aide Memoires, as well as ISRs, and project ratings were duly downgraded to reflect these challenges. Project ratings were



downgraded in response to delays in the land acquisition process, slow physical progress leading to a low disbursement ratio, lack of progress on some key procurement activities, such as the Long-Term Monitoring, Research and Analysis of the Coastal Zone contract, the Community Mobilization and Forestry Contract, and the delay in recruitment of the independent panel of experts in October 2017. However, there were a couple of instances where the ISRs did not provide an explanation for the rating changes. For example, ISR 8 downgraded Overall Implementation Progress to Moderately Unsatisfactory without any explanation. The Aide Memoire during this period recorded detailed evidence of these delays. Once the project got back on track, the ratings were upgraded to reflect the improvement in the pace of implementation.

82. **As noted, the restructurings were less effective than expected in enhancing the project's measurability of results and outcomes, in part because of the World Bank oversight earlier in the project and in part because of the timing and scope of the Client's restructuring requests.** There could have been more extensive support provided to the Client to ensure the timely submission of an official request for the appropriate restructuring of the scope. After January 2020, the Bank's implementation support efforts were intensified, with fortnightly progress review meetings with the PMU and monthly technical field missions. These additional meetings and field visits were effective in terms of resolving critical problems such as the unavailability of construction material and changes in the construction design. Lastly, the project team provided extensive M&E guidance to the Client, including on the proper use of the results framework and the management information system to measure progress and report accordingly.

83. **The Bank provided regular support to the BWDB, both in Dhaka and at project sites, to expedite the payment process for resettlement and land acquisition.** This included supporting the BWDB in establishing a rigorous monitoring mechanism and developing a time-bound action plan to address case-specific land acquisition payment bottlenecks, and arrangements for additional monitoring of land acquisition payments by the BWDB after the closure of the project.

Justification of Overall Rating of Bank Performance

84. **Overall Bank performance is considered Moderately Satisfactory, with minor shortcomings in quality at entry and quality of supervision.** For the most part, the team undertook detailed project preparation as well as regular and effective supervision, including strong collaboration with the PMU. Regular monitoring ensured that CEIP-I met or exceeded most of its revised targets by project closure. However, M&E design weaknesses, capacity issues, land acquisition, and the COVID-19 pandemic posed some challenges for the project.

D. RISK TO DEVELOPMENT OUTCOME

85. **According to the PMU, there is not currently an adequate budget specifically assigned for the O&M of the infrastructure built and rehabilitated under the project, putting its long-term sustainability at risk.** While embankments with protective works, hydraulic structures, and drainage channels were rehabilitated incorporating climate-resilient aspects in their technical design, there are inherent uncertainties in how future climate change impacts will affect the polders' structural integrity as climate projections continue to evolve. However, with the formation of 141 WMGs and 10 WMAs, the communities are overseeing the polders' overall operation and minimal upkeep during cyclones to ensure their overall sustainability (such as closing the gates to avoid tidal flooding and salinity intrusion). Currently, there is no follow-on operation that the Bank is planning exclusively with a coastal resilience agenda. Therefore, there is a high government commitment risk.



86. **The CEIP-I polders are particularly vulnerable to severe riverbank erosion, which poses a direct threat to the integrity of embankments and hydraulic structures.** A potential breach in this critical infrastructure could lead to devastating consequences, especially during natural events such as storm surges and tidal surges. Such a breach would exacerbate the impact of these surges, increasing the area exposed to flooding and salinity intrusion. The infiltration of saline water into the soil not only degrades agricultural land but also undermines the project's sustainability and the livelihoods dependent on it. Addressing these risks requires urgent and robust engineering solutions and management strategies to prevent erosion and strengthen the resilience of the polder infrastructure against the compound threats of natural disasters and environmental changes.

V. LESSONS AND RECOMMENDATIONS

87. **Solid O&M of embankments, hydraulic structures, drainage systems, and afforestation with community mobilization is the foundation of coastal resilience to extract maximum benefits from investments and nurture sustainable interventions.** WMOs should be empowered for the operation and routine minor maintenance of the polders following the experience of CEIP-I. Small works, including minor periodic maintenance of embankment and operation of minor hydraulic infrastructure could be undertaken by the WMOs under a memorandum of understanding with the BWDB. To strengthen the O&M of afforestation activities, a social forestry approach should be used under which the local community is involved in monitoring, maintenance, and management of the plantation with a tripartite agreement between the BWDB, the Bangladesh Forest Department, and social forestry groups.

88. **To mitigate risks and ensure the long-term sustainability of polder infrastructure, it is crucial to support the government in enhancing its preparedness for the O&M phase.** This can be achieved through the development of practical O&M protocols and the implementation of innovative tools to strengthen asset management practices. Additionally, investing in capacity-building initiatives for BWDB staff and WMOs can significantly improve their O&M capabilities. These elements should be considered for inclusion in future programs to effectively reduce risks associated with the O&M phase. Furthermore, exploring the possibility of incorporating O&M components into works contracts, such as build-maintain type contracts, could be a valuable option for future programs, ensuring a more comprehensive approach to infrastructure management.

89. **The unique landscape and dynamics of the coastal polders require a thorough understanding of how the present system functions and its inherent uncertainties with regards to projected climate change impacts.** The dynamic coast of Bangladesh is subject to a multitude of complex natural phenomena, human intervention, and climate change forces, most of which are not yet well understood. To bridge this knowledge gap, there is a need to continue the comprehensive monitoring and morphological assessment of coastal Bangladesh, which was initiated as part of CEIP-I, under the long-term monitoring, research, and analysis of the Bangladesh coastal zone activities. Moreover, the next phase of coastal investment should include collecting empirical evidence on key coastal processes and issues, such as geomorphology, land subsidence, tectonic effects, erosion, SLR, tidal dynamics, river morphology, shoreline changes, and salinity intrusion. The next phase should also focus on the maintenance and further development and application of the numerical tools developed under CEIP-I for practical applications and new investments in the coastal zone. For example, the hydraulic models developed under CEIP-I could be an excellent basis for developing a more detailed storm surge forecasting system for the coastal zone and the polders (which is currently non-existent in Bangladesh).

90. **There is a critical need to complement infrastructure interventions in polders with nature-based solutions to enhance resilience, effectiveness, and community ownership.** Rehabilitating polders with strengthened embankments, alongside afforestation programs on the slopes of embankments and foreshore areas, involves integrating engineering designs with nature-based solutions to protect against the impacts of climate change. This



approach not only aims to enhance physical infrastructure but also leverages the benefits of ecological systems to provide sustainable, long-term resilience against environmental threats. It also addresses the human dimensions of squatters going back to the embankment slopes by garnering public support for the afforestation program (since they are also beneficiaries) and discourages the practice of cutting trees to construct houses illegally on the slope of the embankment.

91. Investing in bank protective works, despite the higher costs compared to retiring an existing embankment towards the countryside, is critical in addressing riverbank erosion effectively and timely. Bank protective measures, while expensive, offer a direct approach to fortifying riverbanks against erosion without the complexities of land acquisition. In contrast, retiring an embankment involves acquiring new land, a process that is not only time consuming but can also lead to considerable delays in project implementation. During these delays, communities face ongoing risks, with potential severe impacts on livelihoods and assets before the new embankment is even operational. Thus, the requirement for immediate protection and the desire to prevent long-term socioeconomic disturbances justify the upfront expenditure on bank protective works.

92. There is a need to accelerate and strengthen the land acquisition process for the rehabilitation of coastal polders. Bangladesh's current legal land acquisition and compensation process can take at least 12 to 15 months, which poses a challenge when undertaking critical construction of embankment and hydraulic works. To balance the need for urgent infrastructure works while following due process, the Bank should support country-level reforms of the land acquisition process, or at a minimum, agree with the GoB to allow an accelerated process for polder infrastructure given the urgent need to protect these areas against tidal and storm surges.

93. Given the changing climate and dynamic coastal processes of the polders, adaptive design should be the cornerstone of any future investment program for coastal resilience. During CEIP-I implementation, following a dynamic coastal process or large-scale riverbank erosion event, it was necessary in several instances to change the alignment of the retired embankments because an embankment had eroded away. Given the emergency nature of the works and the vulnerability of the people in the coastal region to disasters, a risk-based design of polders could help tie together strategic planning, design, and O&M, further improving the efficiency and effectiveness of coastal interventions in Bangladesh.

94. Effective project restructuring and detailed procurement planning, with an emphasis on climate risks and dedicated contract management, are crucial for optimizing resources and ensuring efficient execution in remote, geographically dispersed areas. Efficient project delivery hinges on timely restructuring, ideally immediately after the MTR if scope adjustments are needed and streamlined Bank review processes for swift internal approval. Comprehensive procurement planning, considering contractor availability, capabilities, local contexts, and subproject-specific requirements, is vital for optimizing resources and implementation, especially in remote areas with dispersed works. Contractors experienced in coastal embankment construction should incorporate seasonality, tidal timing, and potential extreme climate events into work plans and unit rates. Well-trained, dedicated contract management capacity at all levels, separate from procurement, and training on the International Federation of Consulting Engineers (commonly known as FIDIC, acronym for its French name *Fédération Internationale Des Ingénieurs-Conseils*) contracts and management for relevant personnel are critical for timely project delivery.



ANNEX 1. RESULTS FRAMEWORK AND KEY OUTPUTS

A. RESULTS INDICATORS

A.1 PDO Indicators

Objective/Outcome: To increase the area protected in selected polders from tidal flooding and frequent storm surges

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Gross area protected	Hectare(Ha)	0.00	100,800.00	66,012.00	65,657.00
		22-Apr-2013	22-Apr-2013	12-Jan-2020	31-Dec-2023

Comments (achievements against targets):

Target almost fully achieved (99.5% achievement compared to the revised target: 65,657 ha compared to 66,012 ha gross area).

The actual (current) gross area protected is the summation of total polder area multiplied by the ratio of completed length of embankment and end target of embankment for 10 Polders (reduced scope of the Project from earlier 17 Polders).

Information source: BWDB project completion report and other supplementary documents.

This is an original indicator but the target was revised in the January 2020 restructuring from 100,800 ha to 66,012 ha.



Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Direct project beneficiaries from increased resilience to climate change	Number	0.00 22-Apr-2013	760,000.00 22-Apr-2013	724,202.00 12-Jan-2020	720,573.00 31-Dec-2023
Female beneficiaries	Percentage	0.00	50.00	50.00	50.00

Comments (achievements against targets):

Target almost fully achieved (99.5% achievement compared to the revised target: 720,573 people compared to 724,202 people).

The actual (current) direct project beneficiaries are the summation of the total Polder population (Bangladesh Bureau of Statistics [BBS] 2011 Census with assumed 1.4% annual growth rate) multiplied by the ratio of the completed length of the embankment and end target of the embankment for 10 Polders.

Information source: BWDB project completion report and other supplementary documents.

This is an original indicator but the target was revised in the January 2020 restructuring from 760,000 (BBS 2001 Census) to 724,202 (BBS 2011 Census with assumed 1.4% annual growth rate).

Objective/Outcome: To improve agricultural production by reducing saline water intrusion in selected polders

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Increase cropping intensity	Percentage	140.00 22-Apr-2013	180.00 22-Apr-2013	180.00 12-Jan-2020	192.00 31-Dec-2023



Comments (achievements against targets):

Target exceeded (106.67% compared to the target: 192% increase cropping intensity against the target of 180%)

The actual increase cropping intensity is the total gross cropped area (single, double and triple crops) divided by total cultivable land in a cropping season for selected 10 Polders of CEIP-I expressed in % (the number of crops a farmer grows in a given agricultural year on the same field).

Information source: BWDB project completion report and other supplementary documents.

This is an original indicator and no change was made during the restructurings.

Objective/Outcome: To improve the Government of Bangladesh's capacity to respond to an emergency

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Contingent Emergency Appropriation	Text	NA	NA	NA	N/A
		22-Apr-2013	22-Apr-2013	12-Jan-2020	31-Dec-2023

Comments (achievements against targets):

This was not triggered.

A.2 Intermediate Results Indicators

Component: Rehabilitation and Improvement of Polders



Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Length of upgraded embankment	Kilometers	0.00	623.00	408.00	405.67
		22-Apr-2013	22-Apr-2013	12-Jan-2020	31-Dec-2023
Comments (achievements against targets): Target almost fully achieved (99.43% of the revised target): End target revised from 623 km to 408 km proportionally to the revised scope of 10 Polders. Information source: BWDB project completion report and other supplementary documents. This is an original indicator but the target was revised in the January 2020 restructuring from 623 km to 408 km.					
Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Drainage Structures replaced	Number	0.00	0.00	88.00	89.00
		22-Apr-2013	22-Apr-2013	12-Jan-2020	31-Dec-2023
Comments (achievements against targets): Target exceeded (101% achievement compared to the revised target: 89 drainage structures replaced compared to 88). Information source: BWDB project completion report and other supplementary documents.					



This was a new indicator added during January 2020 restructuring. As per initial project design, indicator was “drainage structures replaced/upgraded” this is being separated into two indicators to reflect accurately the specific intervention undertaken. The two indicators are now separated as “drainage structures replaced” and “drainage structures repaired”.

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Drainage Structures Repaired	Number	0.00	0.00	8.00	8.00
		22-Apr-2013	22-Apr-2013	12-Jan-2020	31-Dec-2023

Comments (achievements against targets):

Target partially achieved (100% achievement compared to the revised target: 8 compared to 8 drainage structures repaired).

Information source: BWDB project completion report and other supplementary documents.

This was a new indicator added during January 2020 restructuring. As per initial project design, indicator was “drainage structures replaced/upgraded” this is being separated into two indicators to reflect accurately the specific intervention undertaken. The two indicators are now separated as “drainage structures replaced” and “drainage structures repaired”.

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Flushing Inlets replaced	Number	0.00	0.00	80.00	85.00
		22-Apr-2013	22-Apr-2013	12-Jan-2020	31-Dec-2023



Comments (achievements against targets):

Target achieved (106.3% achievement compared to the revised target: 85 compared to 80 flushing inlets replaced).

Information source: BWDB project completion report and other supplementary documents.

This was a new indicator added during January 2020 restructuring. As per initial project design, indicator was “flushing inlets upgraded” this is being separated into two indicators to reflect accurately the specific intervention undertaken. The two indicators are now separated as “flushing inlets replaced” and “flushing inlets repaired”.

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Flushing inlets Repaired	Number	0.00	0.00	44.00	46.00
		22-Apr-2013	22-Apr-2013	12-Apr-2020	31-Dec-2023

Comments (achievements against targets):

Target exceeded (104.5% achievement compared to the revised target: 46 compared to 44 flushing inlets replaced).

Information source: BWDB project completion report and other supplementary documents.

This was a new indicator added during January 2020 restructuring. As per initial project design, indicator was “flushing inlets upgraded” this is being separated into two indicators to reflect accurately the specific intervention undertaken. The two indicators are now separated as “flushing inlets replaced” and “flushing inlets repaired”.

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
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Length of Drainage Channels upgraded	Kilometers	0.00 22-Apr-2013	794.00 22-Apr-2013	305.00 12-Jan-2020	305.85 31-Dec-2023
Comments (achievements against targets): Target achieved (100.3% achievement compared to the revised target: 305.85 km compared to 305 km of drainage channel upgraded). Information source: BWDB project completion report and other supplementary documents. End target revised from 794 km to 305 km proportionally to the revised scope and actual requirements during January 2020 restructuring.					
Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Length of River Bank Protection Works	Kilometers	0.00 22-Apr-2013	0.00 22-Apr-2013	9.37 12-Jan-2020	9.85 31-Dec-2023
Comments (achievements against targets): Target exceeded (105% achievement compared to the revised target: 9.85 km compared to 9.37 km of bank protection). Information source: BWDB project completion report and other supplementary documents. Proposed new indicator "River Bank Protection" was part of the original project scope, but not reflected in the original results framework. BWDB project completion report also refers to additional 7.905 km of precautionary protection works as River Bank Protection Works with total 17.75 km. However, since 9.85 km was done with hard material as dumping material with uniform design standards, the ICR only considered 9.85 km as River Bank Protection Works.					



Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Length of Slope Protection	Kilometers	0.00	0.00	28.00	29.09
		22-Apr-2013	22-Apr-2013	12-Jan-2020	31-Dec-2023
Comments (achievements against targets): Target exceeded (104% achievement compared to the revised target: 29.09 km compared to 28 km of slope protection). Information source: BWDB project completion report and other supplementary documents. Proposed new indicator "Slope Protection" was part of the original project scope, but not reflected in the original results framework.					
Component: Implementation of Social and Environmental Management Frameworks and Plans					
Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Area Afforested	Hectare(Ha)	0.00	300.00	600.00	700.00
		22-Apr-2013	22-Apr-2013	12-Jan-2020	31-Dec-2023
Comments (achievements against targets): Target exceeded (233.33% achievement compared to the revised target: 700 ha compared to 600 ha of afforestation).					



Information source: BWDB project completion report and other supplementary documents.

End target was revised from 300 (ha) to 600 (ha) due to actual requirements during January 2020 restructuring.

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Water Management Association (WMA)	Number	0.00	4.00	10.00	10.00
		22-Apr-2013	22-Apr-2013	12-Jan-2020	31-Dec-2023

Comments (achievements against targets):

Target achieved (100% achievement compared to the revised target: 10 compared to 10 WMAs).

Information source: BWDB project completion report and other supplementary documents.

Description of indicator was revised from “Water Management Organization (WMO)” to “Water Management Association (WMA)” during January 2020 restructuring. End target was also revised from 4 to 10. As per institutional requirement, each polder now has one WMA, there are 10 polders, accordingly, end target revised to 10. As per the initial project design, this was to be piloted only in a few polders, later formation of WMA required for all polders.

Component: Construction Supervision, Project Monitoring and Evaluation, and Coastal Zone Monitoring

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Improved coastal monitoring	Text	Limited data	2.00	2.00	Completed



		22-Apr-2013	22-Apr-2013	12-Jan-2020	31-Dec-2023
Comments (achievements against targets): Target achieved (100% achievement compared to the revised target: 2 compared to 2 studies). 1.Modelling of the long-term physical processes and other relevant phenomena (subsidence, climate change, river morphology etc.) (scale: macro, meso and micro) 2.An investment plan describing a phased polder improvement roadmap for Bangladesh Coast Information source: BWDB project completion report and other supplementary documents. End target proposed to be revised reflecting the scope of works of the analytics (long-term monitoring, research and analysis of the coastal zone).					

Component: Project Management, Technical Assistance, Training and Strategic Studies

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
BWDB Person Days of training provided	Number	0.00	160.00	1,200.00	1,200.00
		22-Apr-2013	22-Apr-2013	12-Jan-2020	31-Dec-2023
Comments (achievements against targets): Target achieved (100% achievement compared to the revised target: 1200 compared to 1200 person days). Information source: BWDB project completion report and other supplementary documents.					



Description of the indicator was revised from “BWDB days of training provided” to “BWDB Person Days of training provided”. The end target was revised from 160 to 1200 reflecting the actual quantity of the training.

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Grievance Redress Committee (GRC)	Number	0.00	17.00	36.00	36.00
		22-Apr-2013	22-Apr-2013	12-Jan-2020	31-Dec-2023

Comments (achievements against targets):

Target achieved (100% achievement compared to the revised target: 36 compared to 36 GRCs).

Information source: BWDB project completion report and other supplementary documents.

End target was revised from 17 to 36 reflecting the number of GRCs formed and functional during January 2020 restructuring. Originally, this was to be one GRC in each polder and in total 17. As per the institutional requirement, there were 36 GRCs formed and functional in 10 Polders.

Indicator Name	Unit of Measure	Baseline	Original Target	Formally Revised Target	Actual Achieved at Completion
Detailed design of future 7 Polders (including EIA, RAP/LAP)	Number	0.00	0.00	7.00	7.00
		22-Apr-2013	22-Apr-2013	12-Jan-2020	31-Oct-2023

Comments (achievements against targets):



Target achieved (100% achievement compared to the revised target: 7 compared to detailed design of future 7 Polders, including EIA, RAP/LAP).

Information source: BWDB project completion report and other supplementary documents.

The Project's scope was reduced from 17 to 10 polders due to the financing constraints. The remaining 7 polders would be moved to a potential second phase of the Project together with additional polders under design, the source of financing of the second phase is not yet determined. However, the detailed design of these 7 polders was ready as well as EIA, RAP/LAP are at final stage of preparation. This was part of the project scope, but was not reflected in the original results framework.



B. KEY OUTPUTS BY COMPONENT

Objective/Outcome 1: Increased area protected in selected polders from tidal flooding and frequent storm surges, which are expected to worsen due to climate change	
Outcome Indicators	1. Gross area protected (1000 x ha) 2. Direct project beneficiaries from increased resilience to climate change (number: 1000 x person) of which female (percentage)
Intermediate Results Indicators	1. Length of upgraded embankment (km) 2. Area afforested (ha) 3. Length of riverbank protection (km) 4. Length of slope protection (km) 5. Improved coastal monitoring (studies) 6. BWDB person days of training provided (number) 7. Grievance redress committee (GRC) (number) 8. Detailed design of 7 future polders (including EIA, RAP/LAP)
Key Outputs by Component (linked to the achievement of the Objective/Outcome 1)	1. 405.67 km of embankment upgraded. 2. 700 ha of area afforested. 3. 9.85 km of riverbank protection works completed. 4. 29.09 km of slope protection completed. 5. Improved coastal monitoring reports prepared. 6. 1,200 person days of training provided. 7. 36 GRCs formed and functional. 8. Detailed design of 7 future polders prepared.
Objective/Outcome 2: Improved agricultural production by reducing saline water intrusion in selected polders	
Outcome Indicators	1. Increase cropping intensity



Intermediate Results Indicators	1. Drainage structures replaced (number) 2. Drainage structures Repaired (number) 3. Flushing inlets replaced (number) 4. Flushing inlets repaired (number) 5. Length of drainage channel upgraded (km) 6. Water management associations (WMAs) (number)
Key Outputs by Component (linked to the achievement of the Objective/Outcome 2)	1. 89 drainage structures replaced. 2. 8 drainage structures repaired. 3. 85 flushing inlets replaced. 4. 46 flushing inlets repaired. 5. 305.85 km of drainage channel upgraded. 6. 10 WMAs formed and registered.
Objective/Outcome 3: Improved capacity of the Government of Bangladesh to respond promptly and effectively to an eligible crisis or emergency	
Outcome Indicators	Contingent Emergency Appropriation Triggered, if requested [Y/N]



ANNEX 2. BANK LENDING AND IMPLEMENTATION SUPPORT/SUPERVISION

A. TASK TEAM MEMBERS

Name	Role
Preparation	
Maria Sarraf	Task Team Leader
Masood Ahmad	Advisor/Overall guidance
Abedalrazq F. Khalil	Water Specialist and Co-Task Team Leader
Tanvir Hossain	Procurement Specialist
Mohammad Reaz Uddin Chowdhury	Financial Management Specialist
Chaohua Zhang	Social Specialist
Anna Charlotte O'Donnell	Team Member
Nadia Sharmin	Environment Specialist
Md. Akhtaruzzaman	Social Specialist
Susmita Dasgupta	Team Member
Manush Hristov	Legal
Junxue Chu	Disbursement
Supervision/ICR	
Swarna Kazi, Ignacio M. Urrutia Duarte	Task Team Leaders
Marc S. Forni	Task Team Leader
A.N.M. Mustafizur Rahman	Procurement Specialist
Hasib Ehsan Chowdhury	Financial Management Specialist
Sk Towhidur Rahaman	Team Member
Mohammad Abdul Basit	Team Member
Olivia Moutusi Sarker	Team Member



Mohammad Rafiqul Islam	Team Member
S M Mehedi Hasan	Team Member
Matheus Van Ledden	Team Member
Debashish Paul Shuvra	ICR Main Contributor
Shourov Kumar Sharma	Procurement Team
Md. Akhtaruzzaman	Social Specialist
S. M. Zulkernine	Environmental Specialist
Esra Arikan	Environmental Specialist
Amani Haque	Team member
Md. Towshikur Rahman	Team Member
Md Kamruzzaman	Procurement Specialist
Josefo Tuyor	Environmental Specialist

B. STAFF TIME AND COST

Stage of Project Cycle	Staff Time and Cost	
	No. of staff weeks	US\$ (including travel and consultant costs)
Preparation		
FY12	28.730	94,641.38
FY13	68.238	413,087.58
FY14	2.900	3,976.80
FY15	0	0.00
Total	99.87	511,705.76
Supervision/ICR		
FY14	64.474	462,016.32
FY15	82.547	347,725.99
FY16	24.777	199,502.07
FY17	40.430	273,754.10



FY18	30.035	232,283.92
FY19	71.820	417,251.72
FY20	74.079	1,084,616.62
FY21	46.575	797,180.20
FY22	42.980	477,299.21
FY23	38.979	533,089.39
FY24	25.024	171,584.97
Total	541.72	4,996,304.51

**ANNEX 3. PROJECT COST BY COMPONENT**

Components	Amount at Approval (US\$M)	Actual at Project Closing (US\$M)	Percentage of Approval
Rehabilitation and Improvement of Polders	291.00	215.56	74%
Implementation of Social and Environmental Management Frameworks and Plans	56.00	49.28	88%
Construction Supervision, Project Monitoring and Evaluation of the Project, and Coastal Zone Monitoring	32.00	36.61	114%
Project Management, Technical Assistance, Training, and Strategic Studies	21.00	10.38	49%
Contingent Emergency Response	0	0.0	NA
Total	400.00	311.84	78%



ANNEX 4. EFFICIENCY ANALYSIS

1. **Scope.** An economic analysis was conducted to assess the rate of return of the capital investments of two components—A. Rehabilitation and Improvement of Polders and B. Implementation of Social and Environmental Management Frameworks and Plans. The total cost of Components A and B was US\$264.84 million, or 85 percent of the total project cost of US\$311.84 million.

2. Assumptions

- The economic life of the infrastructure built under the project is 33 years after the end of the project, based on the assumption for the economic analysis at appraisal.
- The discount rate used at appraisal was 10 percent. This analysis assumed a range of discount rates from 6 to 12 percent.
- The operation and maintenance costs after project completion are 3 percent of the project component costs, as assumed at appraisal.

The following assumptions were used for the Monte Carlo simulations:

- The probability of an extremely severe cyclone like Sidr hitting the project area was assumed to be 2.9 percent at appraisal. This analysis assumed the probability to vary between 2.9 percent and 6.0 percent per year.
- The probability of a flood in Kharif season was assumed to be 67 percent at appraisal. This analysis assumed the probability to vary as 67 ± 25 percent.
- The percent of crop land affected was assumed to be 65 percent at appraisal. This analysis assumed the probability to vary as 65 ± 25 percent.
- The percent of a crop lost on affected cropland was assumed to be 64 percent at appraisal. This analysis assumed the probability to vary as 64 ± 25 percent.
- The analysis at appraisal estimated that 600 people died from Cyclone Sidr in one polder. This analysis assumed the lives saved from a storm like Cyclone Sidr to be between 0.1 percent to 0.3 percent of the population living in the project area.

3. **Costs.** The actual costs of Components A and B were calculated for each year from the beginning to the completion of the project. Table 4.1 shows the costs of each component.

Table 1: Component costs of B. Cyclone Risk Mitigation Infrastructure, in US\$ millions

Component/Year	2016	2017	2018	2019	2020	2021	2022	2023	Total
A	3.0	8.9	26.3	48.4	29.3	17.6	21.5	60.5	215.6
B	2.9	12.8	3.3	15.7	2.7	2.4	8.1	1.3	49.3
A+B	5.9	21.7	29.7	64.0	32.0	20.0	29.6	61.9	264.8

Source: The World Bank calculations for efficiency analysis.

Note: A. Rehabilitation and Improvement of Polders; B. Implementation of Social and Environmental Management Frameworks and Plans.

4. **Methodology.** The implementation completion and results report methodology is tied to the methodology used in the Project Appraisal Document (PAD). The PAD only analyzed the benefits and costs of two polders in detail. This



analysis considered all 10 polders. Some of the data used at appraisal for the benefit calculations were not available at completion. On the other hand, the statistical value of lives saved by the project was not included in the analysis at appraisal and is included in this analysis. Details of the benefit calculation are described below.

5. **Discount rate.** Based on the descriptive methodology, the discount rate is usually set at 12 percent for the economic analysis of World Bank projects. The discount rate used at appraisal was 10 percent. A more recent technical note recommends a different approach of calculating the discount rate as the product of the per capita growth rate of the country and the elasticity of marginal utility of consumption.¹⁰ The per capita annual growth rate of Bangladesh is expected to be 6 percent over the next 20 years. The elasticity of marginal utility of consumption is harder to calculate but is expected to be in the range of 1 to 2. Therefore, a discount rate ranging from 6 to 12 percent was used for the economic analysis at project completion.

6. **Value of statistical life.** One of the benefits of the investments in polders and afforestation is the saving of human lives. A cost-benefit analysis is particularly useful in the comparison of alternatives. To enable decision-making in the context of life-saving investments, some numerical estimates must be attached to the value of life. The relevant concept is the so-called value of a statistical life (VSL). VSL estimates are derived from willingness to pay surveys and reflect the intrinsic value of life to a greater degree than approaches that depend on estimates of foregone income (that is, what the victims could have earned in the labor market if they had survived). The use of VSL mitigates, to a certain extent, the concerns around trivializing the value associated with human life (which has been an obstacle to attributing such values in economic analysis). In addition, although it embodies the individual respondents' utility preferences, the average VSL (as used here) is a neutral measure that is applicable regardless of income earning potential. Due to the paucity of contextualized and reliable VSL estimates applicable for Bangladesh, the benefits transfer method was used to value the lives potentially saved by the project as calculated by Viscusi and Masterman (2017).¹¹ The benefits transfer calculation involves selecting a relevant reference country, in this case, the United States. Viscusi and Masterman provide compelling empirical evidence for the unit income elasticity of the VSL. They estimate the VSL for 189 countries using World Bank data. The estimate of the VSL for Bangladesh made by Viscusi and Masterman was updated with the most recent per capita gross domestic product statistic for Bangladesh and the United States. This estimate was used to arrive at a VSL of US\$1.1 million.

7. **Uncertainties.** Cyclones are uncertain events of nature. Climate change is likely to make them more uncertain in the future. This analysis involves other considerations such as changes in agricultural output, and damage to housing infrastructure. Similarly, the exact number of deaths to be prevented by the project was unknown. To take these uncertainties into consideration this analysis used a range of parameters rather than point estimates. The Monte Carlo simulation process then randomly selects the parameters within those predefined ranges.

8. **Benefits.** This analysis considered avoided damages from cyclones like Sidr. In practice, the rehabilitated polders will protect the project area from all other storms and cyclones that are of lesser strength and magnitude than Sidr. Additionally, two types of benefits included at the appraisal stage were not included because of insufficient data. These benefits were avoided damages to fisheries and roads within the boundaries of the polders. As a result, the measured benefits are underestimates of the actual benefits.

¹⁰ World Bank. 2016. *Discounting Costs and Benefits in Economic Analysis of World Bank Projects*. OPSPQ, May 9, 2016. Washington DC.

¹¹ Viscusi, W., and C. Masterman. 2017. "Income Elasticities and Global Values of a Statistical Life." *Journal of Benefit-Cost Analysis*, 8(2), 226-250. doi:10.1017/bca.2017.12.



9. **Improved crop production because of better drainage.** The project's investment in the polder infrastructure was expected to allow for multiple crops in a larger share of cultivated land in the project area. The cultivation of multiple crops on the same land is measured by cropping intensity. For example, the analysis at appraisal estimated the cropping intensity to be 129 percent for polder 32 at the beginning of the project. The cropping intensity for all of the project area in the year 2022–23 was 192 percent. It is assumed the 63 percent increase in cropping intensity in the project area was due to the project investments. Based on these assumptions, the annual average benefits from crop production were estimated to be US\$54 million.

10. **Avoided crop damage.** In addition to the higher agricultural productivity benefits described above, the rehabilitated and improved polders provide protection to crops in case of storm surges and floods. Three probabilistic parameters were used to estimate the expected avoided annual crop damage in case of a flood hitting the project area.

- a) The probability of a flood in Kharif season when crops are in the field
- b) The percent of cropland affected by the flood
- c) The percent of crop lost on the affected cropland

Based on these probabilities the avoided crop damage was estimated to be between US\$5.6 million and US\$63.0 million.

11. **Avoided livestock damage.** A large number of livestock were killed by Cyclone Sidr. This analysis looked at the percent of deaths of cattle, goats, sheep, and poultry from Cyclone Sidr and more recent floods to estimate the number of livestock that would be saved by the project in case of a similar storm in the project area. The expected annual average value of livestock saved was between US\$0.3 million and US\$0.6 million.

12. **Avoided house damage.** The analysis at appraisal estimated that all the houses in the polders were either completely or partially damaged by Cyclone Sidr. A similar method was used on the updated number of households and the pucca, semi-pucca, and kuccha houses in the project area. The expected annual average avoided damages to houses were between US\$1.6 million and US\$3.3 million.

13. **Afforestation benefits.** The analysis at appraisal calculated the value of non-timber forest products as measured benefits of the afforestation subcomponent. However, due to the unavailability of data, this indicator of afforestation benefits could not be calculated. This analysis considered the carbon sequestration benefits of afforestation using carbon sequestration data from a joint study by the FAO and the World Bank.¹² The annual average benefits from carbon sequestration were estimated to be between US\$0.3 million and US\$2.7 million.

14. **Avoided deaths.** The analysis at appraisal estimated that there were 600 deaths from Cyclone Sidr in one of the polders. This analysis assumed the lives saved from a storm like Sidr to be between 0.1 percent to 0.3 percent of the population living in the project area. The annual average expected benefits from avoided deaths were estimated to be between US\$23.6 million and US\$146.4 million.

15. **The results matrix.** The economic efficiency matrix consists of three indicators, the economic internal rate of

¹²Mukul, Sharif A. 2022. *Summary of Carbon Storage and Sequestration potential through Coastal Afforestation Program under CEIP-I*. mimeo.



return (EIRR), the net present value (NPV), and the benefit-cost ratio (BCR). These indicators were calculated using the assumptions, costs, and benefits described above. The calculations were repeated a thousand times each for the high and low discount rates in the Monte Carlo simulations. The descriptive statistics of the Monte Carlo simulations for the project are shown in Table 4.2.

Table 2: Results Matrix for Economic Indicators of Efficiency

	EIRR	NPV		BCR	
Discount Rate	NA	12.0%	6.0%	12.0%	6.0%
At appraisal:	22.0%	US\$398.6M		Not calculated	
Expected value	25.7%	US\$372.8M	\$1,177.5M	2.9	5.1
Standard deviation	0.03	107.6	263.6	0.5	0.9
Minimum	18.8%	US\$149.3M	US\$595.7M	1.8	3.1
Maximum	33.8%	US\$725.8M	US\$2,004.9M	4.7	8.1
Coefficient of variance	0.11	0.29	0.22	0.19	0.2
Probability of low* outcome	0.0%	0.0%	0.0%	0.0%	0.0%

Source: Table 6.10 of the CEIP-I PAD and The World Bank calculations for efficiency analysis.

Note: *low: < 12% EIRR, < 0 NPV, <1 BCR

The expected values of the economic indicator matrix of EIRR, NPV, and BCR are in the high range. The EIRR for the whole project was 25.7 percent. The NPV was in the range of US\$372.8 to 1,177.5 million and the BCR was in the range of 2.9 to 5.1 for the discount rates of 12 percent and 6 percent, respectively. Note that the NPV at appraisal was calculated using a 10 percent discount rate. Thus, it is not surprising that the NPV at appraisal was within the range calculated in this analysis. Despite the differences in coverage and methodology, the economic indicators at project completion are similar to those at appraisal.

Partial Rating for Economic Analysis: High



ANNEX 5. BORROWER, CO-FINANCIER AND OTHER PARTNER/STAKEHOLDER COMMENTS

The CEIP-I Project Management Unit reviewed the draft implementation completion and results report (ICR). The following comments were provided by the Borrower:

Overall ICR

1. We value the thorough and balanced ICR which extensively covers all aspects of the CEIP-I, highlighting "Lessons Learnt" and the project's impact on physical, environmental, social, and financial dimensions. The Government of Bangladesh (GoB) demonstrated strong commitment and leadership, marking CEIP-I as a key initiative for ongoing disaster risk management efforts.
2. The project achieved long-term disaster preparedness that included construction of new disaster-resilient embankments with build-back-better elements and improved agricultural production by increased cropping intensity. CEIP-I had a bottom-up approach towards building the long-term capacity of the BWDB by focusing on improved coastal monitoring.
3. Overall, the document is factual and gives an overview and account of key milestones, accomplishments, outputs, challenges, and way forward they were addressed.

Achievement of Project Development Objectives (PDO):

4. There exists a strong correlation between the PDO and Bangladesh's 8th five-year plan and Delta Plan 2100, which emphasize continued investment in reducing and mitigating the effects of the underlying risks of coastal polders as well as sustainable development in coastal polders that is resilient to climate change.

Ratings

5. We disagree with the "Moderately Satisfactory" rating given the project's accomplishments. We would argue that the rating can be improved to "Satisfactory" as the project achieved all of its revised key PDO indicator targets, and in the case of cropping intensity (PDO Indicator 3), surpassed the original target as well as had 98.12% disbursement as of project closing.
6. Despite some operational delays influenced by unforeseen events like COVID-19 and severe weather (a number of cyclones), the project has built significant resilience among affected communities. For example, during Cyclone Remal on May 29, 2024, the CEIP-I polders provided a higher level of protection compared to non-CEIP-I polders. CEIP-I embankments withstood storm surges without breaching, preventing storm water from entering the polders, while pluvial flood water drained through the newly-built hydraulic structures.
7. The ongoing land acquisition issues should not downgrade the overall success, as efforts to resolve these are actively monitored and coordinated with relevant offices.
8. The World Bank's role in guiding the project through identification, preparation, and supervision phases was effective, supported by constructive feedback from implementation support missions and timely support on critical



issues. The field visits by the World Bank provided objective and useful feedback for remedial measures. On vital procurement and contractual issues, the concurrence of the Bank was received expeditiously.

Lessons Learned

9. CEIP-I had detailed arrangements in the Resettlement Action Plan and Process Framework to address the adverse impacts of resettlement and land acquisition for the project. However, the major lesson learned from CEIP-I is the delay on land acquisition process and the delayed payment to affected persons. The recommendations regarding accelerated land acquisition procedures for emergency situations need to be strictly enforced under future coastal projects.

10. In addition to the lesson on accelerating and strengthening land acquisition procedures, the BWDB established good practices in addressing resettlement payments to squatters (non-titled affected persons) and gender issues, despite the limitation in terms of project resources.

11. We would like to emphasize the exceptional environmental safeguards compliance demonstrated by the BWDB in preparing site-specific safeguards instruments, periodic reporting, and implementing management plans with training and outreach activities in the complex environment of the coastal polders. This achievement is further complemented by the afforestation activities, which serve as a nature-based solution to mitigate river erosion, enhance natural barriers against storm surges, and promote biodiversity.

12. The implementation period of the rehabilitation of polders project is almost always longer than initially envisaged (e.g., 10-year implementation period for ECRRP), and this was certainly the case here. Should the World Bank be more realistic about the time needed for the projects, which typically not only help rehabilitate polders but also strengthen long-term resilience?



ANNEX 6. SUPPORTING DOCUMENTS (IF ANY)

1. World Bank. 2013. Bangladesh - Coastal Embankment Improvement Project - Phase I (CEIP-I) (P128276). Washington, DC: World Bank.
2. World Bank. 2020. Bangladesh - Coastal Embankment Improvement Project - Phase I (CEIP-I) (P128276): restructuring. Washington, D.C. World Bank Group.
3. World Bank. 2022. Bangladesh - Coastal Embankment Improvement Project - Phase I (CEIP-I) (P128276): restructuring. Washington, D.C. World Bank Group.
4. World Bank. 2023. Bangladesh - Coastal Embankment Improvement Project - Phase I (CEIP-I) (P128276): restructuring. Washington, D.C. World Bank Group.
5. World Bank. *Implementation Status and Results Reports (ISRs)*. Sequence Numbers 1 to 20.
6. World Bank. Aide Memoires from 2012 through 2023.
7. Bangladesh Water Development Board (BWDB). 2016 and 2018. *Baseline Survey Reports*.
8. BWDB. 2022. *Mid-Term Evaluation Report*.
9. BWDB. Annual, Monthly, and Quarterly Progress Reports (DDCS&PMS, M&E, and LTM Consultancy services) from 2014 through 2023.
10. BWDB 2023. *CEIP-I Final Impact Evaluation Report*.
11. BWDB 2024. *CEIP-I Project Completion Report*.